

THE LEAST MODULUS OF A DISTINCT COVERING SYSTEM IS AT MOST 14,500

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ABSTRACT. A covering system is a finite collection of congruences $a_i \pmod{d_i}$ whose union is $\mathbb{Z}$. Erdős asked whether the least modulus of a covering system with distinct moduli can be arbitrarily large. Hough showed that it cannot, with the bound 10^{16} , and Balister, Bollobás, Morris, Sahasrabudhe and Tiba lowered the bound to 616,000. We prove that congruences with distinct moduli, all at least 14,501, never cover the integers, so that every covering system with distinct moduli has least modulus at most 14,500.

The proof keeps the distortion method of Balister et al. and replaces its second-moment estimate for the loss at each prime by a comparison theorem: for each of the distorted measures and every convex nondecreasing φ , the expectation of φ of the number of progressions containing a point is at most its value for aligned progressions under an explicit tilted product measure. This extends Rogers’ theorem on sieving. It bounds the loss at each prime by an expectation over a random smooth number, which we evaluate exactly, up to explicitly bounded truncations, by a certified computation in directed rounding. The computation has been cross-checked by independent code and refereed by an AI agent. The whole proof, including a separate evaluation of the bounds, has also been formally verified in Lean 4 with Mathlib, modulo one `native_decide` evaluation of a numerical checker written in Lean.

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1. INTRODUCTION

1.1. Covering systems and the minimum modulus problem. A *covering system* is a finite collection of arithmetic progressions $a_1 + d_1\mathbb{Z}, \dots, a_k + d_k\mathbb{Z}$ whose union is $\mathbb{Z}$. It is *distinct* if the moduli $d_1, \dots, d_k$ are pairwise distinct. For example,

$$0 \bmod 2, \quad 0 \bmod 3, \quad 1 \bmod 4, \quad 5 \bmod 6, \quad 7 \bmod 12$$

is a distinct covering system with least modulus 2. Covering systems were introduced by Erdős in 1950 [6], who asked whether there are distinct covering systems whose least modulus is arbitrarily large; this is the *minimum modulus problem*. Explicit constructions push the least modulus up to 25 (Gibson [9]), 40 (Nielsen [18]) and 42 (Owens [19]), the largest value known to occur.

In the other direction, Filaseta, Ford, Konyagin, Pomerance and Yu [7] showed that the sum of the reciprocals of the moduli of a distinct covering system grows with its least modulus, and in 2015 Hough [11] resolved the minimum modulus problem negatively: every distinct covering system has least modulus at most 10^{16} . Balister, Bollobás, Morris, Sahasrabudhe and Tiba [1] (henceforth BBMST) introduced the *distortion method*, which gives a simpler proof of Hough’s theorem together with a much better bound: congruences with distinct moduli, all at least 616,000, do not cover $\mathbb{Z}$ [1, Theorem 8.1]. The same method proves Schinzel’s conjecture and gives lower bounds for the density of the uncovered set [1], and it has been used for related questions: the Erdős–Selfridge problem for squarefree moduli [2]; covering systems with squarefree moduli, where the least modulus is at most 118 [4]; bounds for the j -th smallest modulus of a minimal covering system [12] (see also [4]); covering systems whose reciprocal sums are close to 1 [8]; and covering systems of polynomial rings over finite fields, of global function fields and of number fields [15, 16, 23, 17].

1.2. The main results.

Theorem 1.1. *Let $d_1, \dots, d_k$ be distinct integers, all at least 14,501, and let $a_1, \dots, a_k \in \mathbb{Z}$. Then there is an integer x with $x \not\equiv a_i \pmod{d_i}$ for every i .*

Corollary 1.2. *Every covering system with distinct moduli has least modulus at most 14,500. In particular, the largest least modulus of a distinct covering system lies between 42 and 14,500.*

The previous record was 616,000 [1]. The proof of Theorem 1.1 is computer-assisted, in the limited and precise sense explained in Section 1.5. Its numerical part has been certified with outward rounding and cross-checked by independently written code, and the new parts of the proof and of the computation have been checked by an independent referee (an AI agent). Moreover, Theorem 1.1, with the whole of its proof, has been formally verified in Lean 4 with Mathlib, modulo one `native_decide` evaluation of a numerical checker written in Lean (Section 6.8). The formal proof evaluates the per-prime bounds in the same way, for the primes up to $2 \cdot 10^9$, and replaces the explicit prime-counting estimates behind [1, Theorem 6.1] by a sharpened elementary tail estimate (Lemma 4.8). An earlier and independent formalization, with a simpler evaluation and a checker of its own, verified the following weaker statement; its certificate is described in detail in Section 6.4.

Theorem 1.3. *Congruences with distinct moduli, all at least 16,000, do not cover $\mathbb{Z}$. Hence every covering system with distinct moduli has least modulus at most 15,999.*

1.3. Overview of the method. Let $\mathcal{A}$ be a finite family of progressions with distinct moduli, and let $p_1 < p_2 < \dots < p_n$ be the primes dividing the lcm of the moduli. The distortion method of BBMST exposes the progressions prime by prime: at level i it removes the set B_i covered by the progressions whose modulus has largest prime factor p_i . Writing $\mathbb{Z}_{Q_i} = \mathbb{Z}_{Q_{i-1}} \times \mathbb{Z}/p_i^{\gamma_i}$, every point x of $\mathbb{Z}_{Q_{i-1}}$ has a *fibre*, and $\alpha_i(x)$ denotes the proportion of the fibre covered by B_i . Starting from the uniform measure, BBMST build probability measures $P_1, P_2, \dots$; the measure P_i is obtained from P_{i-1} by moving mass inside each fibre from the covered to the uncovered points, but never multiplying the mass of a point by more than $\nu_i = 1/(1 - \delta_i)$, where $\delta_i \in [0, 1)$

is a parameter. The mass that stays on the covered points is the *loss*

$$(1.1) \quad P_i(B_i) = \frac{\mathbb{E}_{P_{i-1}}[(\alpha_i - \delta_i)^+]}{1 - \delta_i},$$

and the progressions do not cover $\mathbb{Z}$ as soon as $\sum_i P_i(B_i) < 1$.

BBMST bound the loss by the first and second moments of α_i , using $(\alpha - \delta)^+ \leq \alpha^2/(4\delta)$, the union bound $\alpha_i \leq U$, where U is a weighted count of the progressions through a point, and a bound on the P_{i-1} -measure of a single progression. The quadratic majorant is exact only at $\alpha = 2\delta$, and it charges heavily covered fibres in proportion to α^2 .

Our main tool is a *comparison theorem* (Theorem 3.2). The distorted measure P_{i-1} is a product of conditional kernels, each bounded by ν_j times the uniform density. We show that for every measure of this kind, every nonnegative combination U of progressions and every convex nondecreasing φ ,

$$\mathbb{E}_{P_{i-1}}[\varphi(U)] \leq \mathbb{E}[\varphi(U^{\text{al}})],$$

where U^{al} is obtained by moving all progressions so that they pass through one point, and by evaluating the resulting count at a random point whose p_j -adic valuations V_j are independent with the *tilted* law $\mathbb{P}(V_j \geq a) = \min(1, \nu_j p_j^{-a})$. Alignment maximizes the overlap of the progressions; for the uniform measure and $\varphi(u) = (u - 1)^+$ the statement is Rogers' theorem on sieving (Section 3.5).

We apply the comparison theorem to the hinge $\varphi(u) = (u - t)^+$ with $0 \leq t \leq \delta_i$. Together with a pointwise inequality (Lemma 4.1) and an enlargement of the family to all admissible moduli, which is where the distinctness of the moduli enters (Lemma 4.3), it gives the bound (Theorem 4.4)

$$(1.2) \quad P_i(B_i) \leq \frac{\mathbb{E}[(U_p(s) - t)^+]}{1 - t}, \quad U_p(s) = \sum_{d|s} w_p(d), \quad w_p(d) = \sum_{j \geq 1: dp^j \geq m} p^{-j},$$

where $p = p_i$, m is the least modulus, and $s = \prod_{q < p} q^{v_q}$ is a random $(p - 1)$ -smooth number with independent exponents, $\mathbb{P}(v_q \geq a) = \min(1, \nu_q q^{-a})$. The right-hand side depends only on m , p and the parameters δ_q , not on the progressions.

It remains to show that, for $m = 14,501$ and a suitable choice of the δ_q , the sum over all primes of the right-hand side of (1.2) is less than 1. For every prime $p \leq 10^9$ we evaluate it exactly, up to truncations whose effect is bounded explicitly (Section 5): for $p < m/2$ by an identity that splits the primes below p into three ranges and reduces the expectation to an enumeration and a two-dimensional law, and for $p > m/2$ by a factorization of the divisor count of s . A certified program with directed rounding shows that the total for $p \leq 10^9$ is at most 0.999927, and a criterion of BBMST handles the larger primes (Section 6). The formal proof uses the same evaluation for $p \leq 2 \cdot 10^9$ together with an elementary tail estimate; for $m = 16,000$ a cruder evaluation suffices, and that version was formalized first. Figure 1 shows the running totals. With the distortion parameters of the certificate for $m = 16,000$, the moment bounds of BBMST would exhaust the whole budget at $p = 193$.

1.4. What is new. The comparison theorem extends a classical inequality, and we are careful about what is prior art.

- *Rogers' theorem* [10, pp. 242–244] states that, for fixed moduli, the density of the integers avoiding the progressions is largest when all the residues are equal. It was proved independently by Simpson [20, Lemma 2.3] and was revisited by Tao [22], whose compression argument has the same structure as our one-coordinate lemma plus induction. Lewis

[14] and Filaseta et al. [7] cite it. It is the case $\nu \equiv 1$, unit weights and $\varphi(u) = (u - 1)^+$ of Theorem 3.2.

- *Filaseta and Kalogirou* [8] used Rogers’ theorem (via Lewis) in the undistorted stages ($\delta = 0$) of a distortion argument, together with the fourth-moment bound $(\alpha - \delta)^+ \leq 27\alpha^4/(256\delta^3)$ for the loss. This is prior art for the fractional-moment bounds of Section 5.8.
- *BBMST’s moment lemma* [1, Lemma 3.6] is the monomial case $\varphi(u) = u^k$ of Theorem 3.2, and [1, Lemma 3.4] is its single-progression case (Section 3.6).

What is new is the extension in two directions at once: to measures given by ν -bounded product kernels, which is what makes the inequality applicable to the distorted measures and produces the tilted law; and to the whole increasing convex order, which allows the non-polynomial hinge $(u - t)^+$. We found no earlier use of convex-order or tilted comparison inequalities to bound the distortion losses themselves. Two further, minor novelties are the exact evaluation of the per-prime bounds (Lemmas 5.1 and 5.2), which is elementary, and an elementary tail estimate (Lemmas 4.7 and 4.8) based on Chebyshev’s bound for the central binomial coefficient, which can replace [1, Theorem 6.1] and Dusart’s estimate for the k -th prime when the margin is not too small.

1.5. What is computer-assisted, and what has been checked. Theorem 4.9 and Proposition 5.4 reduce Theorems 1.1 and 1.3 to finitely many inequalities between explicit numbers: the sum over the primes $p \leq P_{\max}$ of upper bounds c_p for the expectations in (1.2), together with a term for the larger primes, must be less than 1. The computer evaluates these finitely many quantities (for 50,847,534 primes in the case of Theorem 1.1) in IEEE double precision with directed rounding, so that the printed totals are rigorous upper bounds (Section 6.1). No search over covering systems is involved. The distortion parameters were found by floating-point searches, but they are simply inputs: any choice in $[0, \frac{1}{2}]$ gives a valid bound.

The certificate for Theorem 1.1 has been cross-checked by independently written code, with no violation at any prime (Section 6.5). An independent referee (an AI agent) checked the new evaluation by hand and by brute force, audited the rounding and the truncations, and recomputed every per-prime bound up to 10^9 with separately written programs; it found the sum of the exact bounds to lie in $[0.99992148081, 0.99992282962]$ and found no errors. The proofs of Theorems 1.1 and 1.3, each including a separate numerical checker written in Lean, have been formally verified (Section 6.8): the only step that is not checked by Lean’s kernel is the compiled evaluation of the checker, which takes about ten minutes for Theorem 1.1. The certificate for Theorem 1.3 has also been reproduced by independent code. Table 1 summarizes the status of all results of this paper.

1.6. Organization. Section 2 recalls the distortion method. Section 3 states and proves the comparison theorem and relates it to Rogers’ theorem and to the lemmas of BBMST. Section 4 derives the bound on the loss at one prime and reduces the theorems to a finite computation. Section 5 explains how the bound is evaluated, and Section 6 describes the certified computations, their independent verification and the formalization. Section 7 discusses which steps of the method are sharp and where the remaining slack lies. The appendices list the distortion schedules, the budgets by prime ranges, the rounding conventions of the checkers and instructions for reproducing the computations.

2. THE DISTORTION METHOD

This section restates the part of the framework of BBMST [1] that we use, in their notation. We give short proofs, since we use the facts for parameters $\delta_i \in [0, 1)$ rather than $[0, \frac{1}{2}]$, and

TABLE 1. The results of this paper and how they have been checked. “Certified”: a computation with outward rounding; “cross-checked”: recomputed by independently written code; “refereed”: checked by an independent AI agent. No result has yet been refereed by human experts.

theorem	statement	status
1.1	least modulus $\leq 14,500$	certified, cross-checked, refereed; formally verified in Lean 4 (one <code>native_decide</code>)
1.3	least modulus $\leq 15,999$	certified, cross-checked; formally verified in Lean 4 (one <code>native_decide</code> ; an earlier, independent formalization with its own checker)

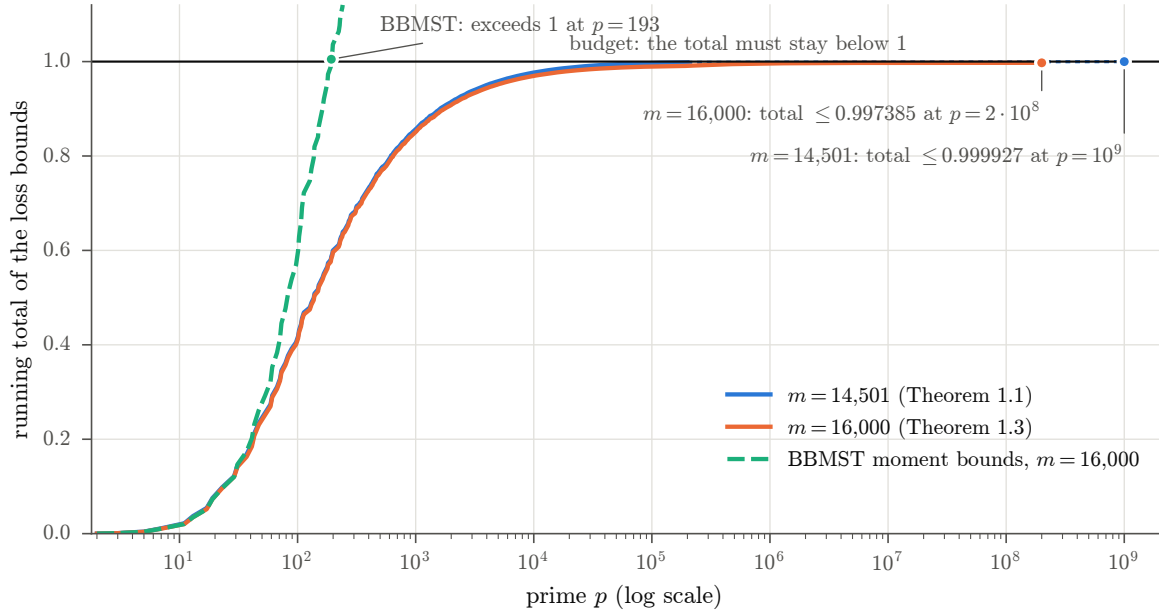


FIGURE 1. Running totals of the certified per-prime loss bounds c_p for the certificates of Theorem 1.1 ($m = 14,501$, blue) and of Theorem 1.3 ($m = 16,000$, orange), for all primes up to $2 \cdot 10^5$; beyond, the orange curve shows every 20,000-th prime, and the dotted blue segment joins the value at $2 \cdot 10^5$ to the certified total at $p = 10^9$. The criterion needs the total to stay below 1. The dashed curve shows the moment bounds $b_p = \min(\mathbb{E}[U_p], \mathbb{E}[U_p^2]/(4\delta_p(1 - \delta_p)))$ of BBMST at $m = 16,000$, computed from the same certified first and second moments and with the same distortion parameters; their total exceeds 1 at $p = 193$ and reaches about 2.2 at $p = 2 \cdot 10^5$.

since the kernel structure of Lemma 2.7 is only implicit in [1]. Results of [1] are cited with the numbering of the arXiv version (arXiv:1811.03547).

2.1. Setting. Let D be a finite set of integers $d \geq 2$ and let $\mathcal{A} = \{A_d : d \in D\}$ with $A_d = a_d + d\mathbb{Z}$. Since $\mathcal{A}$ is indexed by D , the moduli are distinct. We say that $\mathcal{A}$ *covers* if $\bigcup_{d \in D} A_d = \mathbb{Z}$. Let

$$Q = \text{lcm}(D) = \prod_{j=1}^n p_j^{\gamma_j}, \quad p_1 < p_2 < \cdots < p_n, \quad \gamma_j \geq 1,$$

and $Q_i = \prod_{j \leq i} p_j^{\gamma_j}$ for $0 \leq i \leq n$, so $Q_0 = 1$ and $Q_n = Q$. We write $\mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}$. A set $S \subseteq \mathbb{Z}$ is Q_i -*measurable* if it is a union of residue classes modulo Q_i ; we identify such sets with subsets of $\mathbb{Z}_{Q_i}$. Let $X_j = \mathbb{Z}/p_j^{\gamma_j}\mathbb{Z}$. By the Chinese remainder theorem,

$$\mathbb{Z}_{Q_i} \cong X_1 \times \cdots \times X_i, \quad x \mapsto (x \bmod p_1^{\gamma_1}, \dots, x \bmod p_i^{\gamma_i}),$$

and we write $x = (x_1, \dots, x_i)$ and $x_{<j} = (x_1, \dots, x_{j-1})$, which is the reduction of x modulo Q_{j-1} . In particular $\mathbb{Z}_{Q_i} = \mathbb{Z}_{Q_{i-1}} \times X_i$; for $x \in \mathbb{Z}_{Q_{i-1}}$ the *fibre* of x is $F(x) = \{(x, y) : y \in X_i\}$.

The moduli are grouped by their largest prime factor:

$$(2.1) \quad N_i = \{d \in D : d \mid Q_i, d \nmid Q_{i-1}\}, \quad B_i = \bigcup_{d \in N_i} A_d;$$

thus N_i is the set of moduli whose largest prime factor is p_i . This is [1, (3)]. The set B_i is Q_i -measurable. Every $d \in D$ lies in exactly one N_i , so $\bigcup_i B_i = \bigcup_{d \in D} A_d$.

2.2. The distorted measures. Fix $\delta_1, \dots, \delta_n \in [0, 1)$ and put

$$\nu_i = \frac{1}{1 - \delta_i} \geq 1.$$

A probability measure on $\mathbb{Z}_{Q_i}$ is regarded as a measure on $\mathbb{Z}_Q$ by spreading the mass of each residue class modulo Q_i uniformly over the classes modulo Q that it contains [1, footnote 2]. In particular, if P is a measure on $\mathbb{Z}_{Q_{i-1}}$, then $P(x, y) = P(x)/|X_i|$ for $(x, y) \in \mathbb{Z}_{Q_i}$. Let P_0 be the uniform measure. Given P_{i-1} , BBMST define, for $x \in \mathbb{Z}_{Q_{i-1}}$ [1, (4)],

$$(2.2) \quad \alpha_i(x) = \frac{|\{y \in X_i : (x, y) \in B_i\}|}{|X_i|},$$

the proportion of the fibre $F(x)$ covered at level i , and [1, (5)]

$$(2.3) \quad P_i(x, y) = \begin{cases} \max\left(0, \frac{\alpha_i(x) - \delta_i}{\alpha_i(x)(1 - \delta_i)}\right) P_{i-1}(x, y), & (x, y) \in B_i, \\ \min\left(\frac{1}{1 - \alpha_i(x)}, \frac{1}{1 - \delta_i}\right) P_{i-1}(x, y), & (x, y) \notin B_i. \end{cases}$$

The first case occurs only if $\alpha_i(x) > 0$ and the second only if $\alpha_i(x) < 1$, so (2.3) is well defined. BBMST take $\delta_i \in [0, \frac{1}{2}]$. That restriction is used only in their lower bound for the density of the uncovered set [1, Lemma 3.5]; the four facts below hold for $\delta_i \in [0, 1)$ with the same proofs. In our application every δ_i is at most $\frac{1}{2}$ anyway.

Lemma 2.1 ([1, Lemma 2.1]). *For every $x \in \mathbb{Z}_{Q_{i-1}}$ we have $\sum_{y \in X_i} P_i(x, y) = P_{i-1}(x)$. Consequently $P_i(S) = P_{i-1}(S)$ for every Q_{i-1} -measurable set S , and $P_n(S) = P_i(S)$ for every Q_i -measurable set S .*

Proof. Write $\alpha = \alpha_i(x)$ and $\delta = \delta_i$, and recall $P_{i-1}(x, y) = P_{i-1}(x)/|X_i|$. If $\alpha \leq \delta$, the covered points receive nothing and the others receive the factor $1/(1 - \alpha)$, so the fibre has mass $(1 - \alpha) \cdot \frac{1}{1 - \alpha} P_{i-1}(x) = P_{i-1}(x)$. If $\alpha > \delta$, its mass is $(\alpha \cdot \frac{\alpha - \delta}{\alpha(1 - \delta)} + (1 - \alpha) \frac{1}{1 - \delta}) P_{i-1}(x) = P_{i-1}(x)$. Summing over x gives the second statement. If S is Q_i -measurable, it is Q_{j-1} -measurable for every $j > i$, and applying the second statement for $j = i + 1, \dots, n$ gives $P_n(S) = P_i(S)$. $\square$

Lemma 2.2 ([1, Lemma 2.2]). $P_i(x, y) \leq \nu_i P_{i-1}(x, y)$ for every $(x, y) \in \mathbb{Z}_{Q_i}$.

Proof. Both factors in (2.3) are at most $1/(1 - \delta_i)$, because $(\alpha - \delta)/\alpha \leq 1$. $\square$

Lemma 2.3 (the loss; [1, proof of Lemma 3.3]).

$$P_i(B_i) = \frac{1}{1 - \delta_i} \sum_{x \in \mathbb{Z}_{Q_{i-1}}} (\alpha_i(x) - \delta_i)^+ P_{i-1}(x) = \frac{\mathbb{E}_{P_{i-1}}[(\alpha_i - \delta_i)^+]}{1 - \delta_i}.$$

Proof. The fibre $F(x)$ contains $\alpha_i(x)|X_i|$ points of B_i , each of P_{i-1} -mass $P_{i-1}(x)/|X_i|$. By (2.3) they carry together $\max(0, \frac{\alpha_i(x) - \delta_i}{\alpha_i(x)(1 - \delta_i)}) \alpha_i(x) P_{i-1}(x) = \frac{(\alpha_i(x) - \delta_i)^+}{1 - \delta_i} P_{i-1}(x)$ (both sides vanish if $\alpha_i(x) = 0$). Sum over x . $\square$

Proposition 2.4 (the criterion; [1, proof of Theorem 3.1, (13)]). *If $\sum_{i=1}^n P_i(B_i) < 1$, then $\mathcal{A}$ does not cover $\mathbb{Z}$.*

Proof. By Lemma 2.1, $P_n(B_i) = P_i(B_i)$, since B_i is Q_i -measurable. Hence $P_n(\bigcup_d A_d) = P_n(\bigcup_i B_i) \leq \sum_i P_i(B_i) < 1$, so some residue class modulo Q is not covered. $\square$

Remark 2.5 (the rule (2.3) is optimal within each fibre). Fix x with $P_{i-1}(x) > 0$ and let P' be any measure on $F(x)$ with total mass $P_{i-1}(x)$ and $P'(x, y) \leq \nu_i P_{i-1}(x, y)$ for all y ; these are the two properties used in Lemmas 2.1 and 2.2. The uncovered points of $F(x)$ carry at most $\nu_i(1 - \alpha_i(x))P_{i-1}(x)$, so

$$P'(F(x) \cap B_i) \geq \left(1 - \frac{1 - \alpha_i(x)}{1 - \delta_i}\right)^+ P_{i-1}(x) = \frac{(\alpha_i(x) - \delta_i)^+}{1 - \delta_i} P_{i-1}(x),$$

which is the value attained by (2.3). Thus, given P_{i-1} , the rule (2.3) minimizes the loss at level i among all rules with these two properties. (It does not follow that it is optimal for the sum of the losses over all levels; see Section 7.)

2.3. The kernel structure.

Definition 2.6. Let $X = X_1 \times \cdots \times X_r$ be a product of finite nonempty sets and let $\nu_1, \dots, \nu_r \geq 1$. A probability measure μ on X is $(\nu_1, \dots, \nu_r)$ -bounded if there are kernels $K_j: (X_1 \times \cdots \times X_{j-1}) \times X_j \rightarrow [0, \infty)$ such that for all j and all $x_{<j}$,

$$\sum_{y \in X_j} K_j(x_{<j}, y) = 1, \quad K_j(x_{<j}, y) \leq \frac{\nu_j}{|X_j|} \quad (y \in X_j),$$

and $\mu(x) = \prod_{j=1}^r K_j(x_{<j}, x_j)$ for all $x \in X$.

In words: under μ , the conditional law of the j -th coordinate given the previous ones has density at most ν_j with respect to the uniform measure on X_j .

Lemma 2.7. *For $1 \leq i \leq n$, the measure P_{i-1} , viewed as a measure on $\mathbb{Z}_{Q_{i-1}} = X_1 \times \cdots \times X_{i-1}$, is $(\nu_1, \dots, \nu_{i-1})$ -bounded.*

Proof. For $j \leq i - 1$ and $x \in \mathbb{Z}_{Q_{i-1}}$ we have $P_{i-1}(x_{\leq j}) = P_j(x_{\leq j})$ by Lemma 2.1, where $P(x_{\leq j})$ denotes the mass of the residue class of x modulo Q_j . Define

$$K_j(x_{<j}, x_j) = \begin{cases} P_j(x_{<j}, x_j)/P_{j-1}(x_{<j}) & \text{if } P_{j-1}(x_{<j}) > 0, \\ 1/|X_j| & \text{otherwise.} \end{cases}$$

By Lemma 2.1, $\sum_y K_j(x_{<j}, y) = 1$. By Lemma 2.2 and the uniformity of P_{j-1} on fibres, $P_j(x_{<j}, y) \leq \nu_j P_{j-1}(x_{<j}, y) = \nu_j P_{j-1}(x_{<j})/|X_j|$, so $K_j \leq \nu_j/|X_j|$. If $P_{j-1}(x_{<j}) > 0$ for all $j \leq i - 1$, the product $\prod_{j < i} K_j(x_{<j}, x_j)$ telescopes to $P_{i-1}(x)$. Otherwise let j be the least index with $P_{j-1}(x_{<j}) = 0$; then $\prod_{l < j} K_l(x_{<l}, x_l) = P_{j-1}(x_{<j}) = 0$, and $P_{i-1}(x) \leq P_{i-1}(x_{<j}) = P_{j-1}(x_{<j}) = 0$, so both sides vanish. $\square$

2.4. The union bound in the fibre. Every $d \in N_i$ factors uniquely as

$$(2.4) \quad d = m'_d p_i^{j_d}, \quad p_i \nmid m'_d, \quad 1 \leq j_d \leq \gamma_i,$$

and $m'_d \mid Q_{i-1}$. Put $A'_d = \{x \in \mathbb{Z}_{Q_{i-1}} : x \equiv a_d \pmod{m'_d}\}$. In coordinates,

$$(2.5) \quad A'_d = \prod_{j < i} \{x_j \in X_j : x_j \equiv a_d \pmod{p_j^{v_{p_j}(m'_d)}}\},$$

a product of balls in the sense of [Section 3](#).

Lemma 2.8 ([1, proof of Lemma 3.6]). *For every $x \in \mathbb{Z}_{Q_{i-1}}$,*

$$\alpha_i(x) \leq \min(1, U_i(x)), \quad U_i(x) = \sum_{d \in N_i} p_i^{-j_d} \mathbf{1}_{A'_d}(x).$$

Proof. An integer z lies in A_d if and only if $z \equiv a_d \pmod{m'_d}$ and $z \equiv a_d \pmod{p_i^{j_d}}$. Hence the fibre $F(x)$ meets A_d in $p_i^{\gamma_i - j_d}$ points if $x \in A'_d$ and in no point otherwise. Divide by $|X_i| = p_i^{\gamma_i}$ and use the union bound; trivially $\alpha_i \leq 1$. $\square$

2.5. The moment bounds of BBMST. For comparison we record the bounds of [1]. Let $M_i^{(k)} = \mathbb{E}_{P_{i-1}}[\alpha_i^k]$ and $\nu(d) = \prod_{p_j \mid d} \nu_j$ for $d \mid Q$ [1, (8)].

Lemma 2.9 ([1, Lemma 3.3]). *If $0 < \delta_i < 1$, then $P_i(B_i) \leq \min(M_i^{(1)}, M_i^{(2)}/(4\delta_i(1 - \delta_i)))$.*

The first bound follows from $P_i \leq P_{i-1}$ on B_i , the second from [Lemma 2.3](#) and $(\alpha - \delta)^+ \leq \alpha^2/(4\delta)$. BBMST bound the moments via [Lemma 2.8](#) and the estimate $P_{i-1}(b + d\mathbb{Z}) \leq \nu(\gcd(d, Q_{i-1}))/d$ [1, Lemma 3.4], which gives [1, Lemma 3.6]

$$(2.6) \quad M_i^{(k)} \leq \sum_{d_1, \dots, d_k \in N_i} p_i^{-(j_{d_1} + \dots + j_{d_k})} \frac{\nu(\text{lcm}(m'_{d_1}, \dots, m'_{d_k}))}{\text{lcm}(m'_{d_1}, \dots, m'_{d_k})},$$

and [1, Theorem 3.2], valid for $\delta_j \in [0, \frac{1}{2}]$,

$$(2.7) \quad M_i^{(2)} \leq \frac{1}{(p_i - 1)^2} \prod_{j < i} \left(1 + \frac{\nu_j(3p_j - 1)}{(p_j - 1)^2}\right).$$

[Theorem 3.2](#) below reproves (2.6) ([Section 3.6](#)), and [Section 4](#) replaces the second-moment bound by the hinge.

3. THE COMPARISON THEOREM

3.1. Statement. Throughout this section $X = X_1 \times \dots \times X_n$ with $X_i = \mathbb{Z}/q_i^{e_i}\mathbb{Z}$, where $q_i \geq 2$ and $e_i \geq 0$ are integers; in our application the q_i are distinct primes. A *ball of depth a* in X_i ($0 \leq a \leq e_i$) is a set $\{y \in X_i : y \equiv c \pmod{q_i^a}\}$; it has $q_i^{e_i - a}$ elements. A *cylinder* is a product $C = C_1 \times \dots \times C_n$ of balls; we write $a_i(C)$ for the depth of C_i . When the q_i are pairwise coprime, the Chinese remainder theorem identifies the cylinders with the residue classes $c + L\mathbb{Z}$, $L \mid \prod_i q_i^{e_i}$, and then $a_i(C) = v_{q_i}(L)$.

Definition 3.1 (the tilted law). Let $q \geq 2$ be an integer and $\nu \geq 1$. A random variable V with values in $\{0, 1, 2, \dots\}$ has the (q, ν) -tilted law if

$$\mathbb{P}(V \geq a) = \min(1, \nu q^{-a}) \quad (a = 0, 1, 2, \dots).$$

If $\nu \leq q$, this means $\mathbb{P}(V = 0) = 1 - \nu/q$ and $\mathbb{P}(V = a) = \nu q^{-a}(1 - 1/q)$ for $a \geq 1$. For $\nu = 1$ it is the law of the q -adic valuation of a Haar-random q -adic integer.

Theorem 3.2 (comparison theorem). *Let μ be a $(\nu_1, \dots, \nu_n)$ -bounded probability measure on X (Definition 2.6), let $\mathcal{F}$ be a finite family of cylinders with weights $w_C \geq 0$, and let $U = \sum_{C \in \mathcal{F}} w_C \mathbf{1}_C$. Let $V_1, \dots, V_n$ be independent, V_i with the (q_i, ν_i) -tilted law, and define the aligned count*

$$U^{\text{al}} = \sum_{C \in \mathcal{F}} w_C \prod_{i=1}^n \mathbf{1}\{V_i \geq a_i(C)\}.$$

Then for every convex nondecreasing function $\varphi: [0, \infty) \rightarrow \mathbb{R}$,

$$\mathbb{E}_\mu[\varphi(U)] \leq \mathbb{E}[\varphi(U^{\text{al}})].$$

Recall that a random variable A is dominated by B in the *increasing convex order* if $\mathbb{E}\varphi(A) \leq \mathbb{E}\varphi(B)$ for all convex nondecreasing φ for which the expectations exist. Theorem 3.2 says that the law of U under μ is dominated in this order by the law of U^{al} . The name comes from the following description of U^{al} . Translate every cylinder C so that it contains 0; the translate is $C^0 = \{x : q_i^{a_i(C)} \mid x_i \text{ for all } i\}$, and $\mathbf{1}_{C^0}(x) = \prod_i \mathbf{1}\{v_{q_i}(x_i) \geq a_i(C)\}$. Thus U^{al} is the value of the aligned configuration $\sum_C w_C \mathbf{1}_{C^0}$ at a random point whose coordinates have independent valuations with the tilted laws. For $\nu_i = 1$ this random point is uniform, and for $\nu_i > 1$ it is tilted towards 0, where the aligned cylinders overlap the most.

We first prove the theorem for the hinge functions $\varphi(u) = (u - t)^+$, by induction over the coordinates (Sections 3.2 and 3.3); the general case follows because the hinges generate the increasing convex order (Lemma 3.8).

3.2. The one-coordinate lemma. For an integer $N \geq 1$ and a real $\nu \geq 1$ define the *greedy weights*

$$g_{\nu, N}(k) = \min(\nu, \max(0, N - \nu k)), \quad k = 0, 1, \dots, N - 1.$$

Lemma 3.3. *The weights $g = g_{\nu, N}$ are nonincreasing, lie in $[0, \nu]$, and satisfy $\sum_{k < M} g(k) = \min(\nu M, N)$ for $0 \leq M \leq N$. In particular $\sum_{k < N} g(k) = N$. More precisely, with $J = \lfloor N/\nu \rfloor \leq N$, we have $g(k) = \nu$ for $k < J$, $g(J) = N - \nu J \in [0, \nu]$ if $J < N$, and $g(k) = 0$ for $J < k < N$.*

Proof. For $k < J$ we have $k \leq N/\nu - 1$, so $N - \nu k \geq \nu$. For $k > J$ we have $k \geq J + 1 > N/\nu$, so $N - \nu k < 0$. This gives the explicit form, and the partial sums follow: they equal νM for $M \leq J$ and $\nu J + (N - \nu J) = N$ for $M > J$, while $\nu M \leq N$ exactly when $M \leq J$. $\square$

Lemma 3.4 (one coordinate). *Let Y be a finite set with $N = |Y|$, let $\nu \geq 1$, and let $\kappa: Y \rightarrow [0, \nu]$ satisfy $\sum_{y \in Y} \kappa(y) = N$. Let $(B_l)_{l \in L}$ be finitely many subsets of Y with weights $w_l \geq 0$, and put*

$$F(y) = \sum_l w_l \mathbf{1}_{B_l}(y) \quad (y \in Y), \quad G(k) = \sum_l w_l \mathbf{1}\{k < |B_l|\} \quad (0 \leq k < N).$$

Then for every $t \in \mathbb{R}$,

$$\sum_{y \in Y} \kappa(y) (F(y) - t)^+ \leq \sum_{k=0}^{N-1} g_{\nu, N}(k) (G(k) - t)^+.$$

Proof. The proof has three steps.

(a) *Majorization.* For every $s \in \mathbb{R}$,

$$(3.1) \quad \sum_{y \in Y} (F(y) - s)^+ \leq \sum_{k < N} (G(k) - s)^+.$$

Let $S = \{y : F(y) > s\}$ and $M = |S|$. Since $\sum_{k < M} \mathbf{1}\{k < |B_l|\} = \min(M, |B_l|) \geq |S \cap B_l|$,

$$\begin{aligned} \sum_y (F(y) - s)^+ &= \sum_{y \in S} (F(y) - s) = \sum_l w_l |S \cap B_l| - sM \\ &\leq \sum_l w_l \min(M, |B_l|) - sM = \sum_{k < M} (G(k) - s), \end{aligned}$$

and the right-hand side is at most $\sum_{k < N} (G(k) - s)^+$.

(b) *Duality*. For every $\lambda \geq 0$,

$$(3.2) \quad \sum_y \kappa(y) (F(y) - t)^+ \leq \lambda N + \nu \sum_y (F(y) - t - \lambda)^+.$$

Indeed $a^+ \leq \lambda + (a - \lambda)^+$ for every real a and $\lambda \geq 0$; multiply by $\kappa(y) \in [0, \nu]$, sum over y and use $\sum_y \kappa(y) = N$.

(c) *Choice of λ* . Let $h_k = (G(k) - t)^+$. Since G is nonincreasing, so is h . For $\lambda \geq 0$ we have $(G(k) - t - \lambda)^+ = (h_k - \lambda)^+$. Let $J = \lfloor N/\nu \rfloor$ as in Lemma 3.3. If $J = N$, then $\nu = 1$ and $g \equiv 1$; take $\lambda = 0$ in (3.2) and apply (3.1) with $s = t$. If $J < N$, take $\lambda = h_J$. By (3.2) and then (3.1) with $s = t + \lambda$,

$$\begin{aligned} \sum_y \kappa(y) (F(y) - t)^+ &\leq \lambda N + \nu \sum_{k < N} (h_k - \lambda)^+ = h_J N + \nu \sum_{k < J} (h_k - h_J) \\ &= (N - \nu J) h_J + \sum_{k < J} \nu h_k = \sum_{k < N} g(k) h_k, \end{aligned}$$

because $h_k \geq h_J$ for $k < J$ and $h_k \leq h_J$ for $k \geq J$. $\square$

Remark 3.5. The right-hand side of Lemma 3.4 is the maximum of the linear functional $\kappa' \mapsto \sum_k \kappa'_k h_k$ over $\kappa' \in [0, \nu]^N$ with $\sum_k \kappa'_k = N$: since h is nonincreasing, the greedy vector g is optimal (the bathtub principle), and $\lambda = h_J$ is an optimal dual variable. The lemma thus combines two extremal facts: the worst density puts the maximal weight ν on the most covered points, and the worst arrangement of the sets B_l nests them into initial segments of a single ordering of Y .

3.3. Proof of the comparison theorem. We first prove a version for arbitrary product sets, in which only the sizes of the sets enter.

Proposition 3.6 (rectangle form). *Let $X = X_1 \times \cdots \times X_n$ be a product of finite nonempty sets, $N_i = |X_i|$, and let μ be a $(\nu_1, \dots, \nu_n)$ -bounded probability measure on X . Let $(R_c)_{c \in I}$ be finitely many rectangles $R_c = R_{c,1} \times \cdots \times R_{c,n}$ with $R_{c,i} \subseteq X_i$, with weights $w_c \geq 0$, and let $U = \sum_c w_c \mathbf{1}_{R_c}$. Let $\mathbf{k}_1, \dots, \mathbf{k}_n$ be independent random variables with $\mathbb{P}(\mathbf{k}_i = k) = g_{\nu_i, N_i}(k)/N_i$ for $0 \leq k < N_i$. Then for every $t \in \mathbb{R}$,*

$$\mathbb{E}_\mu[(U - t)^+] \leq \mathbb{E}\left[\left(\sum_c w_c \prod_{i=1}^n \mathbf{1}\{\mathbf{k}_i < |R_{c,i}|\} - t\right)^+\right].$$

Proof. Induction on n . For $n = 0$ the space is a point and both sides equal $(\sum_c w_c - t)^+$. Let $n \geq 1$, put $X' = X_1 \times \cdots \times X_{n-1}$ and write points of X as (x', y) with $y \in X_n$. By Definition 2.6, $\mu(x', y) = \mu'(x') K_n(x', y)$, where $\mu'(x') = \prod_{j < n} K_j(x_{<j}, x_j)$ is a $(\nu_1, \dots, \nu_{n-1})$ -bounded probability measure on X' . Write $R_c = R'_c \times R_{c,n}$.

Fix $x' \in X'$. As a function of y , $U(x', y) = \sum_c (w_c \mathbf{1}_{R'_c}(x')) \mathbf{1}_{R_{c,n}}(y)$ is a nonnegative combination of the sets $R_{c,n} \subseteq X_n$, and $\kappa = N_n K_n(x', \cdot)$ takes values in $[0, \nu_n]$ and sums to N_n . Lemma 3.4 gives

$$\sum_y K_n(x', y) (U(x', y) - t)^+ \leq \sum_k \mathbb{P}(\mathbf{k}_n = k) (U^{(k)}(x') - t)^+, \quad U^{(k)} = \sum_c (w_c \mathbf{1}\{k < |R_{c,n}|\}) \mathbf{1}_{R'_c}.$$

Multiply by $\mu'(x')$, sum over x' and exchange the finite sums:

$$\mathbb{E}_\mu[(U - t)^+] \leq \sum_k \mathbb{P}(\mathbf{k}_n = k) \mathbb{E}_{\mu'}[(U^{(k)} - t)^+].$$

For each k , $U^{(k)}$ is a nonnegative combination of rectangles in X' , so by the induction hypothesis $\mathbb{E}_{\mu'}[(U^{(k)} - t)^+] \leq \mathbb{E}[(\sum_c w_c \mathbf{1}\{k < |R_{c,n}|\} \prod_{i < n} \mathbf{1}\{\mathbf{k}_i < |R_{c,i}|\} - t)^+]$. Averaging over k with the weights $\mathbb{P}(\mathbf{k}_n = k)$, and using the independence of $\mathbf{k}_n$ from $(\mathbf{k}_i)_{i < n}$, gives the claim. $\square$

Lemma 3.7 (the tilted law). *Let $N = q^e$, $\nu \geq 1$, and let $\mathbf{k}$ have law $\mathbb{P}(\mathbf{k} = k) = g_{\nu,N}(k)/N$. Define $V = \max\{a \in \{0, \dots, e\} : \mathbf{k} < q^{e-a}\}$. Then $\{V \geq a\} = \{\mathbf{k} < q^{e-a}\}$ and $\mathbb{P}(V \geq a) = \min(1, \nu q^{-a})$ for $0 \leq a \leq e$. Consequently V has the law of $\min(V^\infty, e)$, where V^∞ has the (q, ν) -tilted law, and $\mathbf{1}\{\mathbf{k} < |B|\} = \mathbf{1}\{V \geq a\}$ for every ball B of depth a .*

Proof. The events $E_a = \{\mathbf{k} < q^{e-a}\}$ decrease in a and E_0 is certain, so V is well defined and $\{V \geq a\} = E_a$. By Lemma 3.3, $\mathbb{P}(E_a) = \min(\nu q^{e-a}, N)/N = \min(1, \nu q^{-a})$. The variables V and $\min(V^\infty, e)$ take values in $\{0, \dots, e\}$ and have the same tails there, hence the same law. A ball of depth a has q^{e-a} elements. $\square$

Lemma 3.8 (hinges suffice). *Let A and B be random variables that take finitely many values, all in $[0, \infty)$. If $\mathbb{E}(A - t)^+ \leq \mathbb{E}(B - t)^+$ for every $t \geq 0$, then $\mathbb{E}\varphi(A) \leq \mathbb{E}\varphi(B)$ for every convex nondecreasing $\varphi: [0, \infty) \rightarrow \mathbb{R}$.*

Proof. Let $0 = s_0 < s_1 < \dots < s_r$ be the elements of $\{0\} \cup A(\Omega) \cup B(\Omega)$; if $r = 0$ there is nothing to prove. Let $\sigma_0 = 0$ and $\sigma_j = (\varphi(s_j) - \varphi(s_{j-1})) / (s_j - s_{j-1})$ for $1 \leq j \leq r$. Since φ is nondecreasing and convex, $0 \leq \sigma_1 \leq \dots \leq \sigma_r$. The function

$$\psi(u) = \varphi(0) + \sum_{j=1}^r (\sigma_j - \sigma_{j-1})(u - s_{j-1})^+$$

has slope σ_k on $[s_{k-1}, s_k]$ and $\psi(0) = \varphi(0)$, so $\psi(s_j) = \varphi(s_j)$ for all j . Hence

$$\begin{aligned} \mathbb{E}\varphi(A) &= \mathbb{E}\psi(A) = \varphi(0) + \sum_j (\sigma_j - \sigma_{j-1}) \mathbb{E}(A - s_{j-1})^+ \\ &\leq \varphi(0) + \sum_j (\sigma_j - \sigma_{j-1}) \mathbb{E}(B - s_{j-1})^+ = \mathbb{E}\varphi(B), \end{aligned}$$

because every coefficient $\sigma_j - \sigma_{j-1}$ is nonnegative. $\square$

Proof of Theorem 3.2. Apply Proposition 3.6 to the cylinders $R_C = C$. For each i let $W_i = \max\{a \leq e_i : \mathbf{k}_i < q_i^{e_i-a}\}$ as in Lemma 3.7. Then $\mathbf{1}\{\mathbf{k}_i < |C_i|\} = \mathbf{1}\{W_i \geq a_i(C)\}$, the W_i are independent, and W_i has the law of $\min(V_i, e_i)$. Hence the right-hand side of Proposition 3.6 equals $\mathbb{E}[(\sum_C w_C \prod_i \mathbf{1}\{\min(V_i, e_i) \geq a_i(C)\} - t)^+]$. Since $a_i(C) \leq e_i$, we have $\mathbf{1}\{\min(V_i, e_i) \geq a_i(C)\} = \mathbf{1}\{V_i \geq a_i(C)\}$, and therefore $\mathbb{E}_\mu[(U - t)^+] \leq \mathbb{E}[(U^{\text{al}} - t)^+]$ for every $t \in \mathbb{R}$. Both U and U^{al} take finitely many nonnegative values, and Lemma 3.8 completes the proof. $\square$

3.4. Sharpness.

Proposition 3.9. *Suppose that every cylinder of $\mathcal{F}$ contains 0. Then there is a $(\nu_1, \dots, \nu_n)$ -bounded measure μ for which equality holds in Theorem 3.2 for every φ .*

Proof. The balls of X_i that contain 0 are the sets $q_i^a X_i$, and they are nested. Hence there is an enumeration $\pi_i: \{0, \dots, N_i - 1\} \rightarrow X_i$ such that $\pi_i(\{k : k < q_i^{e_i-a}\}) = q_i^a X_i$ for every a : list 0 first, then the elements of valuation $e_i - 1$, then those of valuation $e_i - 2$, and so on. Let $K_i(x_{<i}, \pi_i(k)) = g_{\nu_i, N_i}(k)/N_i$, independently of $x_{<i}$. Under the product measure μ the vector

$(\pi_i^{-1}(x_i))_i$ has the law of $(\mathbf{k}_i)_i$, and $\mathbf{1}_C(x) = \prod_i \mathbf{1}\{\pi_i^{-1}(x_i) < |C_i|\}$ for $C \in \mathcal{F}$. By Lemma 3.7, U has the law of U^{al} . $\square$

So the comparison theorem loses nothing by itself: the extremal measure puts the maximal density ν_i on the points of highest valuation, and the extremal configuration is the aligned one.

3.5. Rogers' theorem.

Corollary 3.10 (Rogers [10]; Simpson [20]). *Let $d_1, \dots, d_k$ be positive integers, not necessarily distinct, and let $a_1, \dots, a_k \in \mathbb{Z}$. Then the density of $\bigcup_i (a_i + d_i\mathbb{Z})$ is at least the density of $\bigcup_i d_i\mathbb{Z}$. Equivalently, for fixed moduli the density of the integers that avoid all the progressions is largest when all the residues are 0.*

Proof. Let $L = \text{lcm}(d_1, \dots, d_k) = \prod_j q_j^{e_j}$ and $X = \prod_j \mathbb{Z}/q_j^{e_j}$, identified with $\mathbb{Z}_L$; the uniform measure μ is $(1, \dots, 1)$ -bounded, and each progression $a_i + d_i\mathbb{Z}$ is a cylinder C_i of depths $v_{q_j}(d_i)$. Let $U = \sum_i \mathbf{1}_{C_i}$. Since U is integer-valued, the indicator of the union is $\min(1, U) = U - (U - 1)^+$, so the density of the union is $\mathbb{E}_\mu U - \mathbb{E}_\mu (U - 1)^+ = \sum_i 1/d_i - \mathbb{E}_\mu (U - 1)^+$. By Theorem 3.2 with $\varphi(u) = (u - 1)^+$, $\mathbb{E}_\mu (U - 1)^+ \leq \mathbb{E}(U^{\text{al}} - 1)^+$. For $\nu_j = 1$ the tilted variables are the valuations of a uniformly random point, capped at e_j as in Lemma 3.7, so U^{al} has the law of $\sum_i \mathbf{1}_{d_i\mathbb{Z}}$ under the uniform measure, whose union has density $\sum_i 1/d_i - \mathbb{E}(U^{\text{al}} - 1)^+$. $\square$

The direction deserves emphasis. The expected coverage $\mathbb{E}_\mu U = \sum_i 1/d_i$ does not depend on the residues; alignment *maximizes* the overlap $\mathbb{E}(U - 1)^+$ and therefore *minimizes* the covered density $\mathbb{E} \min(1, U)$. For the uniform measure the means of U and U^{al} agree, so Theorem 3.2 is then a statement about the convex order. Rogers' theorem appears in print in Halberstam and Roth [10, pp. 242–244], credited to C. A. Rogers; it was proved independently by Simpson [20, Lemma 2.3] and revisited by Tao [22], whose compression argument, first in one cyclic q -group and then over products, has the same structure as Lemma 3.4 followed by the induction of Proposition 3.6. Lewis [14], Filaseta et al. [7] and Filaseta and Kalogirou [8] use it. Sun [21, Corollary 1.2] proves the analogue $|\bigcup_i a_i G_i| \geq |\bigcup_i G_i|$ for normal Hall subgroups G_i of a finite group, and notes that Tomkinson showed it fails for the Klein four-group. Kucheriaviy [13] shows that the analogue of Rogers' theorem holds in a finite commutative ring exactly when the ring is a product of local rings with linearly ordered ideals. Burchard and Hajaiej [3] prove continuous rearrangement inequalities for supermodular integrands.

3.6. Relation to the lemmas of BBMST. Let $\mu = P_{i-1}$, which is $(\nu_1, \dots, \nu_{i-1})$ -bounded by Lemma 2.7.

A single progression. For a cylinder C and $\varphi(u) = u$, Theorem 3.2 gives $\mu(C) \leq \prod_j \min(1, \nu_j p_j^{-a_j(C)})$. For the progression $b + d\mathbb{Z}$ with $d \mid Q_{i-1}$ this is at most $\nu(d)/d$, which is [1, Lemma 3.4] for the measures P_{i-1} .

Moments. Take $U = U_i$ from Lemma 2.8 and $\varphi(u) = u^k$. Expanding the k -th power,

$$\mathbb{E}[(U_i^{\text{al}})^k] = \sum_{d_1, \dots, d_k \in N_i} p_i^{-(j_{d_1} + \dots + j_{d_k})} \prod_{j < i} \min(1, \nu_j p_j^{-v_{p_j}(L)}) \leq \sum_{d_1, \dots, d_k \in N_i} p_i^{-(j_{d_1} + \dots + j_{d_k})} \frac{\nu(L)}{L},$$

where $L = \text{lcm}(m'_{d_1}, \dots, m'_{d_k})$, since $\max_r v_{p_j}(m'_{d_r}) = v_{p_j}(L)$. Together with $\alpha_i \leq U_i$ this is the moment bound (2.6) of [1, Lemma 3.6]. Thus BBMST's lemma is the monomial case of Theorem 3.2; the theorem makes every convex nondecreasing functional available, and the proof above does not need to compute intersections of progressions.

3.7. Remarks.

- (1) Only the sizes of the sets $R_{c,i}$ enter [Proposition 3.6](#). The balls matter only because nested sets are initial segments of one ordering, which is what makes the description by the tilted law possible ([Lemma 3.7](#)).
- (2) The kernel structure refers to a fixed order of the coordinates, and the induction peels off the last coordinate. For the distorted measures the order is that of the primes $p_1 < p_2 < \dots$, and [Lemma 2.7](#) provides exactly this structure.
- (3) [Lemma 3.4](#) can also be proved for general convex nondecreasing φ by rearrangement: if f^* denotes the decreasing rearrangement of f on $[0, 1]$, the Hardy–Littlewood inequality, the bathtub principle and the Tomić–Weyl majorization inequality give $\int \kappa \varphi(F) \leq \int_0^1 \kappa^* \varphi(F^*) \leq \nu \int_0^{1/\nu} \varphi(F^*) \leq \nu \int_0^{1/\nu} \varphi(G)$, where G is the aligned profile. The argument via duality given above is elementary, and it is the one that has been formalized ([Section 6.8](#)).

4. THE LOSS AT ONE PRIME

In this section $\mathcal{A} = \{A_d : d \in D\}$ is a family of progressions with distinct moduli, all at least $m \geq 2$, and the measures P_i are defined as in [Section 2.2](#). It is convenient to regard the distortion parameters as a *schedule* defined on all primes: a function $q \mapsto \delta_q \in [0, 1)$ with $\delta_i = \delta_{p_i}$ for the primes $p_i \mid Q$, and $\nu_q = 1/(1 - \delta_q)$. For primes $q \nmid Q$ the value δ_q does not affect the measures, but it enters the bounds below.

4.1. The pointwise step.

Lemma 4.1. *Let $0 \leq t \leq \delta < 1$, $\alpha \in [0, 1]$ and $U \geq \alpha$. Then*

$$\frac{(\alpha - \delta)^+}{1 - \delta} \leq \frac{(U - t)^+}{1 - t}.$$

The hypothesis $t \leq \delta$ cannot be dropped.

Proof. The left-hand side is nondecreasing in α , so we may replace α by $\min(1, U)$. If $U \leq \delta$, the left-hand side vanishes. If $\delta < U \leq 1$, both sides are affine functions of U on $[\delta, 1]$ (note $U > t$), they are equal to 1 at $U = 1$, and the left-hand side has the larger slope $1/(1 - \delta) \geq 1/(1 - t)$; hence it is the smaller one on $[\delta, 1]$. If $U > 1$, the left-hand side equals $1 \leq (U - t)/(1 - t)$. If $t > \delta$, the same comparison of slopes shows that the inequality fails for $\alpha = U$ slightly below 1. $\square$

4.2. The random smooth number and the weights.

Definition 4.2. Let p be a prime. The *random $(p - 1)$ -smooth number* is $s = s_p = \prod_{q < p} q^{v_q}$, the product over the primes $q < p$, where the exponents v_q are independent and v_q has the (q, ν_q) -tilted law of [Definition 3.1](#): $\mathbb{P}(v_q \geq a) = \min(1, \nu_q q^{-a})$. For a positive integer d let

$$j_0(d) = \min\{j \geq 1 : dp^j \geq m\}, \quad w_p(d) = \sum_{j \geq j_0(d)} p^{-j} = \frac{p^{1-j_0(d)}}{p-1}, \quad U_p(s) = \sum_{d \mid s} w_p(d).$$

The number $w_p(d)$ is the total weight $\sum p^{-j}$ of the moduli $dp^j \geq m$ that a family could contain at level p with cofactor d . We record some elementary facts, assuming $1 \leq \nu_q \leq q$ for all $q < p$.

- (i) $0 < w_p(d) \leq 1/(p - 1)$, with equality if and only if $dp \geq m$. Hence $U_p(s) \leq \tau(s)/(p - 1)$, where τ is the number of divisors, with equality if $p \geq m$.
- (ii) If $s \mid s'$, then $U_p(s) \leq U_p(s')$.

(iii) For a $(p-1)$ -smooth d , $\mathbb{P}(d \mid s) = \prod_{q \mid d} \nu_q q^{-a} = \nu(d)/d$, where $\nu(d) = \prod_{q \mid d} \nu_q$. Consequently

$$(4.1) \quad \mathbb{E}[U_p(s)] = \sum_d w_p(d) \frac{\nu(d)}{d} = \sum_{j \geq 1} p^{-j} \left(\prod_{q < p} \left(1 + \frac{\nu_q}{q-1} \right) - \sum_{\substack{d < \lceil m/p^j \rceil \\ d \text{ (p-1)-smooth}}} \frac{\nu(d)}{d} \right),$$

since $w_p(d) = \sum_{j \geq 1} p^{-j} \mathbf{1}\{d \geq \lceil m/p^j \rceil\}$ and $\sum_d \nu(d)/d = \prod_{q < p} (1 + \nu_q/(q-1))$; the inner sums are finite and vanish once $p^j \geq m$.

(iv) For $\theta \geq 0$, $\mathbb{E}[\tau(s)^\theta] = \prod_{q < p} \mathbb{E}[(v_q + 1)^\theta]$ with

$$(4.2) \quad \mathbb{E}[(v_q + 1)^\theta] = 1 + \nu_q \left(1 - \frac{1}{q} \right) \sum_{a \geq 1} q^{-a} ((a+1)^\theta - 1), \quad \mathbb{E}[(v_q + 1)^2] = 1 + \frac{\nu_q(3q-1)}{(q-1)^2}.$$

The second formula is the factor of [1, Lemma 3.7].

Facts (i) and (ii) are immediate. For (iv), $\mathbb{E}[(v+1)^\theta] = (1 - \nu/q) + \sum_{a \geq 1} \nu q^{-a} (1 - 1/q)(a+1)^\theta$, and $\nu/q = \nu(1 - 1/q) \sum_{a \geq 1} q^{-a}$; for $\theta = 2$ use $\sum_{a \geq 1} x^a(a^2 + 2a) = x(3-x)/(1-x)^3$ with $x = 1/q$.

4.3. Enlargement. For $d \in N_i$ recall the factorization $d = m'_d p_i^{j_d}$ of (2.4).

Lemma 4.3 (enlargement). *Let $p = p_i$ and let s' be a positive integer all of whose prime factors are less than p . Then*

$$\sum_{d \in N_i} p^{-j_d} \mathbf{1}\{m'_d \mid s'\} \leq U_p(s').$$

Proof. The map $d \mapsto (m'_d, j_d)$ is injective on N_i , because $d = m'_d p_i^{j_d}$. This is where the distinctness of the moduli is used: the family contains at most one progression for each pair (m', j) . For $d \in N_i$ we have $m'_d p_i^{j_d} = d \geq m$ and $j_d \geq 1$, so $j_d \geq j_0(m'_d)$. Hence the left-hand side is at most $\sum_{m' \mid s'} \sum_{j \geq j_0(m')} p^{-j} = U_p(s')$. $\square$

4.4. The per-prime bound.

Theorem 4.4. *Let $1 \leq i \leq n$ and $p = p_i$. Then for every t with $0 \leq t \leq \delta_i$,*

$$P_i(B_i) \leq \frac{\mathbb{E}[(U_p(s) - t)^+]}{1 - t},$$

where $s = s_p$ is the random $(p-1)$ -smooth number of Definition 4.2.

Proof. By Lemma 2.3, Lemma 2.8 and Lemma 4.1,

$$P_i(B_i) = \frac{\mathbb{E}_{P_{i-1}}[(\alpha_i - \delta_i)^+]}{1 - \delta_i} \leq \frac{\mathbb{E}_{P_{i-1}}[(U_i - t)^+]}{1 - t}, \quad U_i = \sum_{d \in N_i} p^{-j_d} \mathbf{1}_{A'_d}.$$

By Lemma 2.7, P_{i-1} is a $(\nu_1, \dots, \nu_{i-1})$ -bounded measure on $X_1 \times \dots \times X_{i-1}$, and by (2.5) each A'_d is a cylinder with depths $a_j(A'_d) = v_{p_j}(m'_d)$. Apply Theorem 3.2 with $\varphi(u) = (u - t)^+$. Let V_j ($j < i$) be independent with the (p_j, ν_j) -tilted laws and $s' = \prod_{j < i} p_j^{V_j}$; then $\prod_{j < i} \mathbf{1}\{V_j \geq v_{p_j}(m'_d)\} = \mathbf{1}\{m'_d \mid s'\}$, so

$$\mathbb{E}_{P_{i-1}}[(U_i - t)^+] \leq \mathbb{E} \left[\left(\sum_{d \in N_i} p^{-j_d} \mathbf{1}\{m'_d \mid s'\} - t \right)^+ \right] \leq \mathbb{E}[(U_p(s') - t)^+] \leq \mathbb{E}[(U_p(s) - t)^+].$$

The second inequality is Lemma 4.3. For the third, let $s'' = \prod q^{v_q}$ over the primes $q < p$ that do not divide Q , with independent tilted exponents; then $s = s' s''$ has the law of Definition 4.2 and $U_p(s') \leq U_p(s)$ by (ii). $\square$

For $t = \delta_i = 0$ the bound is the first moment $\mathbb{E}[U_p(s)]$. The next lemma allows other parameters.

Lemma 4.5. *For $\alpha \in [0, 1]$, the function $\delta \mapsto (\alpha - \delta)^+/(1 - \delta)$ is nonincreasing on $[0, 1]$. Consequently $P_i(B_i) \leq \mathbb{E}_{P_{i-1}}[(\alpha_i - d)^+]/(1 - d)$ for every $d \in [0, \delta_i]$.*

Proof. Where it is positive, the derivative is $(\alpha - 1)/(1 - \delta)^2 \leq 0$. The second statement follows from Lemma 2.3. $\square$

Corollary 4.6 (moment variants). *Under the hypotheses of Theorem 4.4:*

- (a) $P_i(B_i) \leq \mathbb{E}[U_p(s)]$.
- (b) For $0 < d \leq \delta_i$, $P_i(B_i) \leq \mathbb{E}[U_p(s)^2]/(4d(1 - d))$.
- (c) For $\theta > 1$ and $0 < d \leq \delta_i$,

$$P_i(B_i) \leq \frac{c_\theta(d) \mathbb{E}[U_p(s)^\theta]}{1 - d} \leq \frac{c_\theta(d) \mathbb{E}[\tau(s)^\theta]}{(p - 1)^\theta(1 - d)}, \quad c_\theta(d) = \max_{u \in [d, 1]} \frac{u - d}{u^\theta}.$$

Explicitly, $c_\theta(d) = 1 - d$ if $\theta d \geq \theta - 1$, and $c_\theta(d) = \frac{d}{\theta - 1} \left(\frac{\theta d}{\theta - 1}\right)^{-\theta}$ otherwise.

Proof. (a) is Theorem 4.4 with $t = 0$. For (b) and (c), start from Lemma 4.5. For (b) use $(\alpha - d)^+ \leq \alpha^2/(4d)$, which is $(\alpha - 2d)^2 \geq 0$, then $\alpha_i \leq U_i$ and Theorem 3.2 with $\varphi(u) = u^2$, followed by Lemma 4.3 and fact (ii) as in the proof of Theorem 4.4. For (c) observe that $(\min(1, u) - d)^+ \leq c_\theta(d)u^\theta$ for $u \geq 0$: for $u < d$ the left side vanishes, for $u \in [d, 1]$ this is the definition of c_θ , and for $u > 1$ the left side is $1 - d \leq c_\theta(d) \leq c_\theta(d)u^\theta$. Since $\alpha_i \leq \min(1, U_i)$ and $u \mapsto u^\theta$ is convex and nondecreasing, the same chain gives the first inequality; the second is fact (i). Finally, $g(u) = (u - d)u^{-\theta}$ has $g'(u) = u^{-\theta-1}(\theta d - (\theta - 1)u)$, so g increases up to $u^* = \theta d/(\theta - 1)$ and decreases afterwards; its maximum on $[d, 1]$ is $g(u^*) = \frac{d}{\theta - 1}(u^*)^{-\theta}$ if $u^* \leq 1$ and $g(1) = 1 - d$ otherwise. $\square$

Part (b) is BBMST's second-moment bound, with $M_i^{(2)}$ replaced by the enlarged moment $\mathbb{E}[U_p(s)^2]$; part (c) with $\theta = 4$ and the unconstrained maximum $27/(256d^3)$ in place of $c_4(d)$ is the bound of Filaseta and Kalogirou [8].

4.5. Reduction to a finite computation. The primes beyond a large threshold X are handled with $\delta_q = \frac{1}{2}$ and the following elementary estimate, which replaces [1, Theorem 6.1] and Dusart's lower bound for the k -th prime.

Lemma 4.7 (Chebyshev tail). *Let $X \geq 2^{27}$ be an integer, let S be a finite set of primes, all greater than X , and let $g(q) = 2(3q - 1)/(q - 1)^2$. Then*

$$\sum_{p \in S} \frac{1}{(p - 1)^2} \prod_{\substack{q \in S \\ q < p}} (1 + g(q)) \leq \frac{100}{189 X}.$$

Proof. Primes in a dyadic block. Let $n \geq 2^{27}$ and let T be a set of primes in $(n, 2n]$. The product of the elements of T divides $\binom{2n}{n} \leq 4^n$ and each element exceeds n , so $n^{|T|} \leq 4^n$. Since $n \geq 2^{27}$, this gives $2^{27|T|} \leq 2^{2n}$, that is, $|T| \leq 2n/27$.

The product over a block. For $q > n \geq 4000$ we have $q - 1 \geq n$ and $g(q) = 6/(q - 1) + 4/(q - 1)^2 \leq (6 + 4/n)/n \leq 6.001/n$. Hence $\prod_{q \in T} (1 + g(q)) \leq \exp(|T| \cdot 6.001/n) \leq \exp(12.002/27) \leq \frac{25}{16}$, where the last inequality is a numerical fact ($\exp(0.44452) < 1.5598$).

Summation. Split S into the blocks $S_J = S \cap (X2^J, X2^{J+1}]$, $J \geq 0$, and apply the two previous steps with $n = X2^J \geq 2^{27}$. For $p \in S_J$ the product over $q \in S$, $q < p$, runs over the

blocks $S_0, \dots, S_J$, so it is at most $(25/16)^{J+1}$, while $(p-1)^2 \geq (X2^J)^2$ and $|S_J| \leq 2X2^J/27$. Hence block J contributes at most

$$\frac{2X2^J}{27} \cdot \left(\frac{25}{16}\right)^{J+1} \cdot \frac{1}{(X2^J)^2} = \frac{25}{216X} \left(\frac{25}{32}\right)^J,$$

and summing the geometric series gives $\frac{25}{216X} \cdot \frac{32}{7} = \frac{100}{189X}$. $\square$

Lemma 4.8 (sharpened Chebyshev tail). *In the setting of Lemma 4.7, the sum is at most C_X/X , where $C_X = \frac{7051}{50000}$ if $X \geq 2 \cdot 10^8$, $C_X = \frac{12473}{100000}$ if $X \geq 10^9$, and $C_X = \frac{5941}{50000}$ if $X \geq 2 \cdot 10^9$.*

Outline. The complete proof is formal (`lean/MinModulus/Tail2/`); we outline it. Write $\theta(X, N] = \sum_{X < q \leq N} \log q$ over primes q .

- (1) *Chebyshev with a leftover.* For $10^8 \leq X \leq N$, $\theta(X, N] \leq 2(\log 2 + 10^{-6})(N - X) + \log(4/3)X$. The interval is cut into dyadic pieces from the top, each bounded by $\binom{2M}{M} \leq 4^M$ or $\binom{2M+1}{M} \leq 4^M$ as in the proof of Lemma 4.7, and the remaining piece $(X, N']$ with $N' < 2X$ by $\binom{N'}{X} 3^X \leq 4^{N'}$.
- (2) *Partial summation.* Summation by parts over all integers in $(X, N]$ turns (1) into $\sum_{X < q \leq N} g(q) \leq a \log(\log N / \log X) + b / \log X$ with $a = 12(\log 2 + 10^{-6})(1 + 10^{-7})$ and $b = 6(1 + 10^{-7}) \log(4/3)$.
- (3) *Telescoping.* With $P(x) = \prod_{q \in S, q \leq x} (1 + g(q))$ we have $\prod_{q < p} (1 + g(q)) g(p) = P(p) - P(p^-)$ and $1/(p-1)^2 = g(p)/(2(3p-1))$. Summation by parts on the dyadic cells $[X2^k, X2^{k+1}]$, $0 \leq k < 30$, with a potential built from chords of the convex function $v \mapsto (1 + v/\log X)^a$, bounds the sum over $S \cap (X, X2^{30}]$ by an explicit expression in the values of this function at $v = k \log 2$; these 31 values are bounded in exact rational arithmetic.
- (4) The primes of S above $X2^{30}$ contribute at most $\frac{100}{189} P(X2^{30})/(X2^{30})$, by Lemma 4.7 applied at $X2^{30}$, and $P(X2^{30})$ is bounded by (2). $\square$

Theorem 4.9 (the criterion). *Let $m \geq 2$ and $X \geq 2^{27}$ be integers, and let $q \mapsto \delta_q$ be a schedule with $\delta_q \in [0, \frac{1}{2}]$ for all primes q and $\delta_q = \frac{1}{2}$ for $q > X$. For each prime $p \leq X$ let c_p be a real number such that at least one of the following holds, where $s = s_p$:*

- (i) $c_p \geq \mathbb{E}[(U_p(s) - t)^+]/(1 - t)$ for some $t \in [0, \delta_p]$;
- (ii) $c_p \geq \mathbb{E}[U_p(s)^2]/(4d(1 - d))$ for some $d \in (0, \delta_p]$;
- (iii) $c_p \geq c_\theta(d) \mathbb{E}[\tau(s)^\theta]/((p-1)^\theta(1 - d))$ for some $\theta > 1$ and $d \in (0, \delta_p]$.

Let $T_X = \prod_{q \leq X} (1 + \nu_q(3q - 1)/(q - 1)^2)$. If

$$(4.3) \quad \sum_{p \leq X} c_p + \frac{100}{189} \cdot \frac{T_X}{X} < 1,$$

then no family of congruences with distinct moduli, all at least m , covers $\mathbb{Z}$.

Proof. Let $\mathcal{A}$ be such a family and run the distortion method with $\delta_i = \delta_{p_i}$. All tilts satisfy $1 \leq \nu_q \leq 2 \leq q$. For $p_i \leq X$, Theorem 4.4 and Corollary 4.6 give $P_i(B_i) \leq c_{p_i}$; in particular $c_p \geq 0$ for all $p \leq X$. For $p = p_i > X$ we have $\delta_i = \frac{1}{2}$, and $(u - \frac{1}{2})^+/\frac{1}{2} \leq u^2$ for all $u \geq 0$ (since $u^2 - 2u + 1 \geq 0$), so Theorem 4.4, fact (i) and (4.2) give

$$P_i(B_i) \leq \mathbb{E}[U_p(s)^2] \leq \frac{\mathbb{E}[\tau(s)^2]}{(p-1)^2} = \frac{1}{(p-1)^2} \prod_{q < p} \left(1 + \frac{\nu_q(3q-1)}{(q-1)^2}\right) \leq \frac{T_X}{(p-1)^2} \prod_{X < q < p} (1 + g(q)),$$

because $\nu_q = 2$ for $q > X$. Summing over the primes $p_i > X$, which form a subset of the set S of all primes in $(X, p_n]$, and applying [Lemma 4.7](#) to S , we obtain $\sum_{p_i > X} P_i(B_i) \leq \frac{100}{189} T_X/X$. Therefore $\sum_i P_i(B_i)$ is at most the left-hand side of (4.3), and [Proposition 2.4](#) applies. $\square$

Remark 4.10. By [Lemma 4.8](#), [Theorem 4.9](#) remains true with $\frac{100}{189}$ replaced in (4.3) by the constant C_X of [Lemma 4.8](#) when $X \geq 2 \cdot 10^8$; the proof is the same. The formal proof of [Theorem 1.1](#) uses this with $X = 2 \cdot 10^9$ ([Section 6.3](#)).

Remark 4.11. The schedule and the choice among (i)–(iii) are free parameters: any choice gives a valid criterion. In particular, the numbers δ_q need not be the ones that minimize the left-hand side of (4.3), and they need not be computed exactly, as long as the c_p are upper bounds for the corresponding quantities computed with the δ_q and ν_q actually used (see [Section 6.1](#)).

5. EVALUATING THE BOUND

We now describe how upper bounds c_p for the quantities in [Theorem 4.9](#) are computed. There are two certified computations. The one for [Theorem 1.1](#) ($m = 14,501$, primes up to 10^9) evaluates the comparison bound exactly, up to truncations whose effect is bounded explicitly ([Sections 5.4 to 5.6](#)). The one for [Theorem 1.3](#) ($m = 16,000$, primes up to $2 \cdot 10^8$) uses a cruder evaluation ([Sections 5.7 and 5.8](#)), which was formalized first; the formal proof of [Theorem 1.1](#) evaluates the bounds exactly, for the primes up to $2 \cdot 10^9$ ([Section 6.3](#)). This section describes the mathematics of the computations; the arithmetic, which is carried out with directed rounding, is described in [Section 6](#) and [Appendix C](#).

5.1. Overview. For $m = 14,501$ the checker `certificate-14501/src/rigx.c` treats the primes $p \leq 10^9$ as follows.

- (a) $p < 17$: $\delta_p = 0$, and c_p is an upper bound for the first moment $\mathbb{E}[U_p(s)]$ ([Section 5.3](#)).
- (b) $17 \leq p < m/2$ (921 primes, the last one 7,247): c_p is an upper bound for $\mathbb{E}[(U_p(s) - t)^+]/(1 - t)$ with $t = \delta_p$, computed from the identity of [Lemma 5.1](#).
- (c) $m/2 < p \leq 10^9$: the same quantity, computed from the factorization of [Lemma 5.2](#).

In (b) and (c) the program also computes the first- and second-moment bounds of [Corollary 4.6\(a\),\(b\)](#) and keeps the smallest candidate; in the certified run this was the comparison bound at every prime with $\delta_p > 0$. The costs in (b) are then multiplied by $1 + 2^{-20}$ and those in (c) by $1 + 2^{-13}$. These deliberate safety margins cost $4.6 \cdot 10^{-6}$ in total, and they make the independent checks of [Section 6.5](#) strict. The primes $p > 10^9$ are handled by [Proposition 5.4](#).

For $m = 16,000$ the checker `src/rigcert.c` treats the primes $p \leq 2 \cdot 10^8$ as follows.

- (a') $p < 31$: $\delta_p = 0$ and the first moment.
- (b') $31 \leq p \leq 2 \cdot 10^5$: the smallest of upper bounds for $\mathbb{E}[U_p(s)]$, for $\mathbb{E}[U_p(s)^2]/(4d(1-d))$ with $d = \min(\delta_p, \frac{1}{2})$, and for the comparison bound $\mathbb{E}[(U_p(s) - t)^+]/(1 - t)$ for each t in the list $\{0.05, 0.075, 0.1, \dots\} \cap [0, \delta_p)$ together with $t = \delta_p$, evaluated with the decomposition of [Section 5.7](#). In the final run the comparison bound was the smallest at all 17,974 of these primes, and the minimum over t was always attained at $t = \delta_p$.
- (c') $2 \cdot 10^5 < p \leq 2 \cdot 10^8$: the smallest of the first and second moment bounds and the fractional-moment bounds of [Corollary 4.6\(c\)](#) for twelve values of θ ([Section 5.8](#)).

The primes $p > 2 \cdot 10^8$ are handled by the tail estimate of [Theorem 4.9](#).

5.2. The structure of U_p . Let $N_j = \lceil m/p^j \rceil$ for $j \geq 1$. Then $j_0(d) = \min\{j \geq 1 : d \geq N_j\}$, and $w_p(d) = 1/(p - 1)$ exactly when $d \geq N_1$. Writing $c(d) = 1 - p^{1-j_0(d)}$, which vanishes for

$d \geq N_1$, we get

$$(5.1) \quad (p-1)U_p(s) = \tau(s) - \sum_{d|s, d < N_1} c(d).$$

Thus $U_p(s)$ is essentially a divisor count, corrected by the small divisors of s . If $p \geq m/2$, then $N_1 \leq 2$, so $d = 1$ is the only divisor that can be below N_1 : $U_p(s) = \tau(s)/(p-1)$ if $p \geq m$, and $U_p(s) = (\tau(s) - 1 + 1/p)/(p-1)$ if $m/2 \leq p < m$, since then $p < m \leq p^2$ and $j_0(1) = 2$. For small p the correction in (5.1) is substantial, since N_1 is large.

5.3. The first moment. For the primes with $\delta_p = 0$ we use (4.1): the Euler product $\prod_{q < p} (1 + \nu_q/(q-1))$ is rounded up and the finite sums $\sum_{d < N_j} \nu(d)/d$ are rounded down. The same formula gives the first-moment candidate for larger p . A similar finite formula gives $\mathbb{E}[U_p(s)^2]$ as a ν -weighted sum over pairs of divisors (`src/CERT_NOTES.md`); it is one of the candidates, but it is never the smallest bound in either certified run.

5.4. Exact evaluation for $p < m/2$: the S/B/H identity. Let $p < m/2$, so that $N_1 \geq 3$. Choose an integer y with

$$(5.2) \quad y^2 \geq N_1, \quad y \geq N_2, \quad y \leq N_1$$

(for instance $y = N_1$; the certified run takes $y = N_1$ if $N_1 \leq 23$, and $y = \max(\lceil \sqrt{N_1} \rceil, N_2)$ otherwise). Split the primes $q < p$ into three ranges,

$$S = \{q < y\}, \quad B = \{y \leq q < \min(N_1, p)\}, \quad H = \{N_1 \leq q < p\},$$

write $s = s_S s_B s_H$ accordingly, and put

$$\tau_S = \tau(s_S), \quad T = \prod_{q \in B \cup H} (v_q + 1), \quad F_S = \sum_{d|s_S, d < N_1} c(d), \quad b = \sum_{q \in B, v_q \geq 1} a_q(s_S),$$

where $a_q(s_S) = \#\{d \mid s_S : dq < N_1\}$.

Lemma 5.1 (the S/B/H identity). $(p-1)U_p(s) = \tau_S T - F_S - (1 - \frac{1}{p})b$.

Proof. Since $\tau(s) = \tau_S T$, by (5.1) we must show that $\sum_{d|s, d < N_1} c(d) = F_S + (1 - \frac{1}{p})b$. Let $d \mid s$ with $d < N_1$, and write $d = d_S d_B d_H$ with $d_S \mid s_S$, $d_B \mid s_B$ and $d_H \mid s_H$. Every prime of H is at least N_1 , so $d_H = 1$. Every prime of B is at least y , and $y^2 \geq N_1$, so d_B has at most one prime factor counted with multiplicity: $d_B = 1$, or $d_B = q$ for a prime $q \in B$ with $v_q \geq 1$. The divisors with $d_B = 1$ contribute F_S . For a fixed $q \in B$ with $v_q \geq 1$, there are exactly $a_q(s_S)$ divisors $d = d_S q < N_1$ with $d_S \mid s_S$, and each of them satisfies $d \geq q \geq y \geq N_2$, so that $dp^2 \geq m$, and $dp < m$, since $d < N_1$. Hence $j_0(d) = 2$ and $c(d) = 1 - 1/p$ for each of them. $\square$

The identity reduces the hinge to three independent pieces: s_S , the exponents $(v_q)_{q \in B}$, and s_H . The numbers $a_q(s_S)$ count divisors below $N_1/q \leq N_1/y \leq y$, so the vector $(a_q(s_S))_{q \in B}$, the *profile* of s_S , depends only on the divisors of s_S below y . With $\beta = t(p-1) + F_S + (1 - \frac{1}{p})b$,

$$\mathbb{E}[(U_p(s) - t)^+] = \frac{1}{p-1} \mathbb{E}_{s_S} [\tau_S \mathbb{E}[(T - \beta/\tau_S)^+]],$$

where the inner expectation is over $(v_q)_{q \in B \cup H}$, with s_S fixed. The program evaluates this as follows.

- It enumerates s_S by a depth-first search over the exponents. A subtree (the exponent of the current prime at least a , later exponents free) is pruned when the bound $\mathbb{P} \cdot \tau \cdot \mathbb{E}[(v+1)\mathbf{1}\{v \geq a\}] \cdot \prod_{\text{later}} \mathbb{E}[v+1] \cdot \mathbb{E}[T]$ for its contribution to $\mathbb{E}[\tau_S T]$ is below $10^{-14} \mathbb{E}[\tau_S] \mathbb{E}[T]$, and this bound is then added to the result; this is valid because $(\tau_S T - \beta)^+ \leq \tau_S T$.

- For each profile, it computes the joint law of (T, b) by a dynamic program over the primes of B , started from the law of $T_H = \prod_{q \in H} (v_q + 1)$. The law of T_H is maintained incrementally as p increases; primes enter H from both ends, since N_1 decreases as p increases.
- It evaluates each $\mathbb{E}[(T - x)^+; b]$ from suffix sums of the law of T , by (5.4) below, with x rounded down.

In the certified run the enumeration visited 143,025,319 values of s_S in total; at each level the pruned part was at most $1.95 \cdot 10^{-7}$ and the lost mass of Section 5.6 at most $2.19 \cdot 10^{-6}$ of the level bound. The identity was first used in the independent recomputation of the certificate for Theorem 1.3 (Section 6.5), whose program for these primes makes its own choices of y .

5.5. Exact evaluation for $p > m/2$: the τ -split. Let $p \geq m/2$. By Section 5.2, $(p-1)U_p(s) = \tau(s) - c_1$ with $c_1 = 1 - 1/p$ if $p < m$ and $c_1 = 0$ if $p \geq m$, so that $\mathbb{E}[(U_p(s) - t)^+] = \mathbb{E}[(\tau(s) - X)^+]/(p-1)$ with $X = c_1 + t(p-1)$. Fix $Y \leq p$ (in the certified run $Y = 7,253$, the least prime with $N_1 \leq 2$), and write $\tau(s) = \tau_s \tau_l$ with $\tau_s = \prod_{q < Y} (v_q + 1)$ and $\tau_l = \prod_{Y \leq q < p} (v_q + 1)$, which are independent. For an integer $v \geq 0$ let $f(v)$ be the least integer of the form $2^i 3^j$ that is at least $v + 1$; thus $f(v) = v + 1$ for $v \leq 3$.

Lemma 5.2 (the τ -split). *Let $V \geq \tau_l$ be a random variable that depends only on the exponents v_q , $Y \leq q < p$; for example $V = \prod_{Y \leq q < p} f(v_q)$. Then*

$$\mathbb{E}[(\tau(s) - X)^+] \leq \sum_x \mathbb{P}(V = x) x \mathbb{E}[(\tau_s - X/x)^+].$$

Proof. Since $u \mapsto (u - X)^+$ is nondecreasing and $V \geq \tau_l \geq 1$, we have $(\tau_s \tau_l - X)^+ \leq (\tau_s V - X)^+ = V(\tau_s - X/V)^+$. Take expectations, using the independence of τ_s and V . $\square$

The law of τ_s is computed once, by a dynamic program on $\{1, \dots, 2^{22}\}$, together with the suffix sums that give $\mathbb{E}[(\tau_s - x)^+]$ for every x by (5.4). The program takes $V = 2^{e_2} 3^{e_3}$, obtained from $\prod_{Y \leq q < p} f(v_q)$ by capping e_3 at 2: a factor 3^j that would exceed the cap is replaced by $4^j = 2^{2j}$, which is larger. The law of (e_2, e_3) is computed by a dynamic program over the primes in $[Y, p)$, maintained incrementally in p , and states with $e_2 > 40$ go to the lost mass. Since $\mathbb{P}(v_q \geq 4) = \nu_q q^{-4}$ is tiny for $q \geq Y$, the domination $\tau_l \leq V$ loses very little. At the largest primes, however, the threshold X/V can exceed the table of τ_s , and the program then bounds $\mathbb{E}[(\tau_s - X/V)^+]$ by the lost mass, which is valid but conservative: the referee of Section 6.5 found that this overstates the contribution of the primes above $2 \cdot 10^5$ by about $5 \cdot 10^{-7}$.

5.6. Truncations and lost mass. All laws are computed on finite tables with finitely many exponents. Every dynamic program stores upper bounds for the probabilities of the cells it keeps, and the mass that leaves the tables is not discarded but accounted for as follows. Let W be the random variable, a product of factors $v_q + 1$, whose law is being built, and suppose that a hinge $\mathbb{E}[(\Lambda W - \beta)^+]$ with $\Lambda, \beta \geq 0$ is to be bounded. Split the probability space into the kept cells and the lost part L . On L we use $(\Lambda W - \beta)^+ \leq \Lambda W$, so that

$$\mathbb{E}[(\Lambda W - \beta)^+] \leq \sum_{\text{kept } w} \mathbb{P}(W = w) (\Lambda w - \beta)^+ + \Lambda \mathbb{E}[W; L].$$

The lost part $E_{\text{lost}} = \mathbb{E}[W; L]$ is accumulated as a sum of nonnegative terms: when a prime q is added,

$$(5.3) \quad E_{\text{lost}} \leftarrow E_{\text{lost}} \mathbb{E}[v_q + 1] + \sum_w \mathbb{P}(W = w) w h(A_w),$$

where $h(A) = \mathbb{E}[(v_q + 1)\mathbf{1}\{v_q \geq A\}] = \nu_q q^{-A}(A + 1 + \frac{1}{q-1})$ and A_w is the least exponent that makes the cell w leave the table. Cells leave the table when the value exceeds the table size or the exponent exceeds the cut, and, in the program for [Theorem 1.1](#), when a cell is negligible (“dust”, $\mathbb{P}(W = w)w < 10^{-30}$), in which case its whole contribution $\mathbb{P}(W = w)w \mathbb{E}[v_q + 1]$ is moved to E_{lost} . The pruned subtrees of [Section 5.4](#) are treated in the same way. This replaces the formula $\mathbb{E}[W] - \sum_w w \mathbb{P}(W = w)$, which is algebraically identical but suffers from cancellation. Finally, the hinge of a kept law is evaluated as a sum of nonnegative terms: with the suffix sums $S_0(j) = \sum_{w \geq j} \mathbb{P}(W = w)$ and $S_1(j) = \sum_{i \geq j} S_0(i)$, and $x' = \lfloor x \rfloor + 1$ for $x \geq 0$,

$$(5.4) \quad \mathbb{E}[(W - x)^+] = \sum_{w \geq x'} (w - x) \mathbb{P}(W = w) = S_1(x' + 1) + (x' - x) S_0(x').$$

5.7. A simpler evaluation for $m = 16,000$. The certificate for [Theorem 1.3](#) evaluates the comparison bound for $31 \leq p \leq 2 \cdot 10^5$ with a decomposition that is cruder but simpler to formalize. Fix p and $t \in [0, \delta_p]$, let $z = z_p$ be the largest prime $q \leq 47$ with $q < \min(p, N_1)$, or $z = 1$ if there is none, and split $s = s_A s_B$ with $s_A = \prod_{q \leq z} q^{v_q}$ and $s_B = \prod_{z < q < p} q^{v_q}$, which are independent. Let $U_A(s_A) = \sum_{d|s_A} w_p(d)$.

Lemma 5.3. $U_p(s) \leq U_A(s_A) + (\tau(s_B) - 1) \tau(s_A) / (p - 1)$.

Proof. Every divisor of s is $d_A d_B$ with $d_A | s_A$ and $d_B | s_B$. The divisors with $d_B = 1$ contribute $U_A(s_A)$. There are $\tau(s_A)(\tau(s_B) - 1)$ others, and each has weight at most $1/(p - 1)$. $\square$

The choice of z puts the primes that can divide short divisors into s_A ; any z gives a valid bound. Let

$$G(k, u) = \mathbb{E}_{s_B} \left[\left(u + \frac{(\tau(s_B) - 1)k}{p - 1} - t \right)^+ \right],$$

which is nondecreasing in u and in k . By [Lemma 5.3](#) and the independence of s_A and s_B , $\mathbb{E}[(U_p(s) - t)^+] \leq \mathbb{E}_{s_A}[G(\tau(s_A), U_A(s_A))]$. We split this expectation according to the size of s_A :

$$(5.5) \quad \mathbb{E}[(U_p(s) - t)^+] \leq \sum_{s_A \leq X_p} \mathbb{P}(s_A) G(\tau(s_A), U_A(s_A)) \\ + \sum_{k \leq K} R_k G\left(k, \frac{k}{p - 1}\right) + \frac{\mathbb{E}[\tau(s_A); \tau(s_A) > K] \mathbb{E}[\tau(s_B)]}{p - 1}.$$

Here the first sum runs over all z -smooth $s_A \leq X_p = \max(10^4, 16N_1^{4.5})$, which are enumerated exactly together with $\mathbb{P}(s_A) = \prod_{q \leq z} \mathbb{P}(v_q = a_q)$, $\tau(s_A)$ and $U_A(s_A)$; $K = 2 \cdot 10^4$; and $R_k \geq \mathbb{P}(\tau(s_A) = k, s_A > X_p)$. For $s_A > X_p$ we used $U_A(s_A) \leq \tau(s_A)/(p - 1)$ and the monotonicity of G in u ; for $\tau(s_A) = k > K$ we used $G(k, k/(p - 1)) = \mathbb{E}[(k\tau(s_B)/(p - 1) - t)^+] \leq k\mathbb{E}[\tau(s_B)]/(p - 1)$. (Terms with $s_A \leq X_p$ and $\tau(s_A) > K$ are counted twice, which is harmless.) The numbers R_k are computed as an upper bound for $\mathbb{P}(\tau(s_A) = k)$, from a dynamic program over the primes $q \leq z$, minus a lower bound for the enumerated mass with $\tau(s_A) = k$. The largest number of enumerated s_A at a single prime was 12,114,518.

The function G is evaluated from the law of $\tau(s_B)$. Put $a = k/(p - 1)$. If $u \geq t$, then, since $\tau(s_B) \geq 1$, $G(k, u) = u - t + (\mathbb{E}[\tau(s_B)] - 1)a$ with $\mathbb{E}[\tau(s_B)] = \prod_{z < q < p} (1 + \nu_q/(q - 1))$. If $u < t$, then $G(k, u) = a \mathbb{E}[(\tau(s_B) - c)^+]$ with $c = 1 + (t - u)/a > 1$, which is evaluated by (5.4). The law of $\tau(s_B)$ is computed by a dynamic program on $\{1, \dots, 10^6\}$, maintained incrementally in p : when p increases, the primes in (z_p, p) that are not yet included are added one at a time; since z_p can only decrease once $p > 47$, and s_B is empty for $p \leq 47$, no prime ever has to be removed. Exponents a with $\mathbb{P}(v_q = a) < 10^{-30}$ or $a > 63$ are dropped, and the lost mass is

accounted for as in [Section 5.6](#). The law of $\tau(s_A)$ and $\mathbb{E}[\tau(s_A); \tau(s_A) > K]$ are computed in the same way, without dropping exponents.

5.8. Fractional moments for $2 \cdot 10^5 < p \leq 2 \cdot 10^8$ ($m = 16,000$). For $p > 2 \cdot 10^5$ we have $p > m$, so $U_p(s) = \tau(s)/(p-1)$ exactly, and [Corollary 4.6\(c\)](#) gives

$$c_p = \min_{\theta} \frac{c_{\theta}(\delta_p) \mathbb{E}[\tau(s)^{\theta}]}{(p-1)^{\theta}(1-\delta_p)}, \quad \theta \in \{\frac{3}{2}, 2, \frac{5}{2}, 3, \frac{7}{2}, 4, 5, 6, 7, 8, 10, 12\}.$$

The products $\mathbb{E}[\tau(s)^{\theta}] = \prod_{q < p} \mathbb{E}[(v_q + 1)^{\theta}]$ are maintained as running products over the primes. Each factor is computed from [\(4.2\)](#): the series is summed until its terms are negligible, and the remainder is bounded rigorously, since for $a \geq A+1$ the terms $b_a = q^{-a}(a+1)^{\theta}$ satisfy $b_{a+1}/b_a \leq r := ((A+3)/(A+2))^{\theta}/q$, so that $\sum_{a > A} b_a \leq b_{A+1}/(1-r)$. Half-integer powers use a square root rounded in the required direction. At the eleven primes of this range examined in the independent recomputation, the minimum is attained at $\theta = \frac{5}{2}$ for $p \leq 10^7$ and at $\theta = 3$ for $p \geq 5 \cdot 10^7$ (`tests/xcheck_final/out/task4_full.out`). The 11,060,953 primes in this range contribute 0.006510785610 in total, about six times the exact values of their comparison bounds ([Section 7.4](#)).

5.9. The tail. [Theorem 4.9](#) with $X \geq 2^{27}$ requires an upper bound for $T_X = \prod_{q \leq X} (1 + \nu_q(3q-1)/(q-1)^2)$, which the programs compute as a running product, rounded up. Note that $T_X = \mathbb{E}[\tau(s)^2]$ for the random number s built from all primes $q \leq X$. For $m = 16,000$ and $X = 2 \cdot 10^8$ the tail term in [\(4.3\)](#) is $\frac{100}{189} T_X/X \leq 0.000849343$. For $m = 14,501$ and $X = 10^9$ it would be about $3.7 \cdot 10^{-4}$, which exceeds the margin $1 - \eta \approx 7.3 \cdot 10^{-5}$; there we use the criterion of BBMST, which the programs check as well. The formal proof of [Theorem 1.1](#) instead takes $X = 2 \cdot 10^9$ and the constant of [Lemma 4.8](#), which make the tail term 0.000047341491 ([Section 6.3](#)).

Proposition 5.4 (via [\[1, Theorem 6.1\]](#)). *In the setting of [Theorem 4.9](#), suppose instead of [\(4.3\)](#) that $\eta := \sum_{p \leq X} c_p < 1$, that $k := \pi(X) \geq 10$, and that*

$$\frac{T_X}{1-\eta} \leq (\log k + \log \log k - 3)^2 k.$$

Then no family of congruences with distinct moduli, all at least m , covers $\mathbb{Z}$.

Proof. Following [\[1, Section 6\]](#), perform a sieve step at every prime, with the trivial step ($B_i = \emptyset$, $P_i = P_{i-1}$) at primes not dividing Q , and write p_i for the i -th prime. With $i_0 = 0$ and $\kappa = 1$, [\[1, \(19\)\]](#) defines $f_k = \mu_k^{-1} \prod_{i \leq k} (1 + \nu_{p_i}(3p_i - 1)/(p_i - 1)^2)$, where $\mu_k = 1 - \sum_{i \leq k} P_i(B_i)$. Their hypothesis [\[1, \(20\)\]](#) for these κ and i_0 is the bound [\(2.7\)](#) of [\[1, Theorem 3.2\]](#), valid since all $\delta_q \leq \frac{1}{2}$ (including factors for primes that do not divide Q only increases the product). By [Theorem 4.4](#) and [Corollary 4.6](#), $\mu_k \geq 1 - \eta > 0$, so $f_k \leq T_X/(1-\eta)$, and [\[1, Theorem 6.1\]](#) applies. Its proof sets $\delta_i = \frac{1}{2}$ for $i > k$ and uses Dusart's bound $p_i \geq i(\log i + \log \log i - 1)$ [\[5\]](#). $\square$

5.10. The distortion schedules. For $m = 14,501$ the schedule is $\delta_q = 0$ for $q < 17$ and, for $q \geq 17$, piecewise linear in $\log q$ through the 103 knots listed in [Appendix A](#), from $(17, 0.00375)$ to $(2 \cdot 10^5, 0.46)$, and constant 0.46 beyond; $\delta_q = \frac{1}{2}$ for $q > 10^9$ in the tail argument. For

$m = 16,000$ it is

$$\delta_q = \begin{cases} 0, & q < 31, \\ 0.05, & 31 \leq q < 50, \\ \text{piecewise linear in } \log q \text{ through the knots of Table 6,} & 50 \leq q \leq 50,000, \\ 0.4092, & q > 50,000, \end{cases}$$

for $q \leq 2 \cdot 10^8$, and $\delta_q = \frac{1}{2}$ for $q > 2 \cdot 10^8$. Both are shown in Fig. 2(a). They were found by numerical searches with floating-point versions of the evaluators, and as noted in Remark 4.11, the certificates do not depend on how they were found. The values actually used by the certified programs differ from these by rounding (Section 6.1).

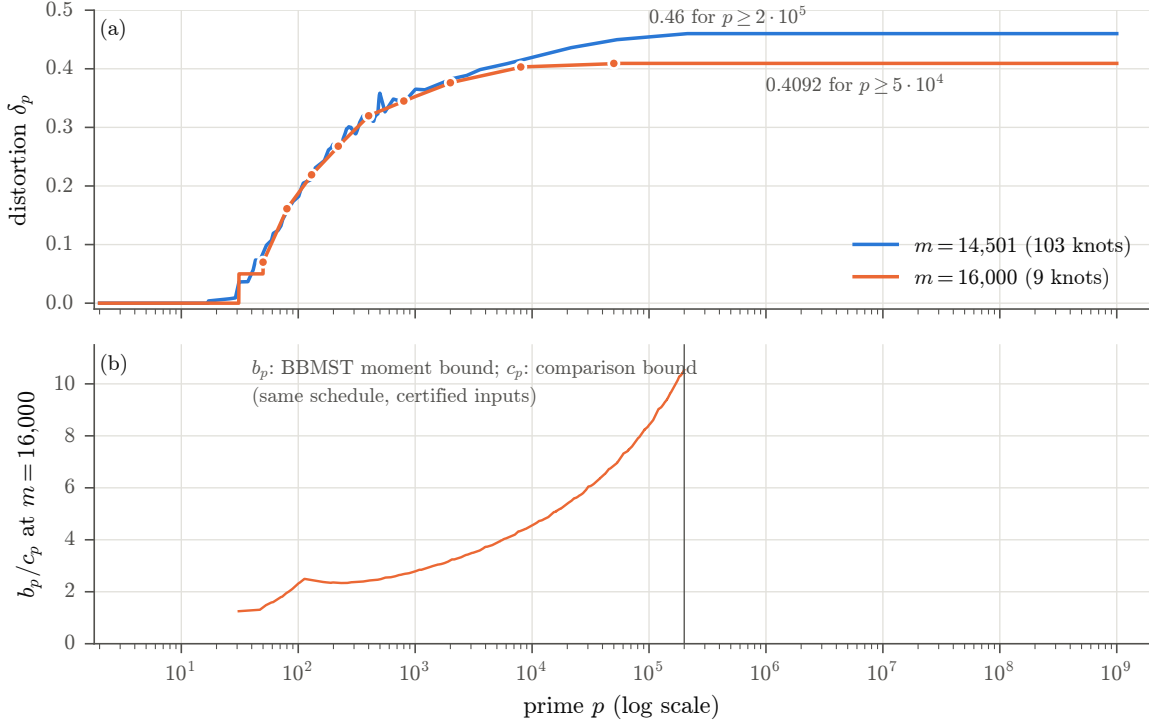


FIGURE 2. (a) The distortion schedules δ_p of the two certified runs; the dots are the knots of the schedule for $m = 16,000$. (b) The ratio b_p/c_p of the moment bound $b_p = \min(\mathbb{E}[U_p], \mathbb{E}[U_p^2]/(4\delta_p(1 - \delta_p)))$ of BBMST to the certified comparison bound c_p at $m = 16,000$, for $31 \leq p \leq 2 \cdot 10^5$, with the same schedule; both are computed from certified quantities.

With the same schedule, the comparison bound is smaller than the moment bound of BBMST by a factor between 1.25 and 1.31 for $31 \leq p < 50$, between 1.48 and 3.22 for $50 \leq p < 2,000$, and between 3.23 and 10.54 for $2,000 \leq p \leq 2 \cdot 10^5$ (Fig. 2(b)).

6. THE CERTIFIED COMPUTATIONS

6.1. Arithmetic with directed rounding. The certified checkers `certificate-14501/src/rigx.c` (for Theorem 1.1) and `src/rigcert.c` (for Theorem 1.3) are single C files that implement Section 5 in IEEE-754 double precision with directed rounding. The first was derived from the second and keeps all of its conventions. (Separate implementations in fixed-point integer

arithmetic are part of the formalizations of [Theorems 1.1 and 1.3](#); see [Sections 6.3 and 6.8](#).) The conventions are as follows; [Appendix C](#) lists the rounding direction of the quantities.

- The processor is switched to round-upward mode once and stays there, so every $+$, $-$, $\times$, $/$ returns an upper bound for the exact result of its operands. Lower bounds are computed as $\text{RD}(x \circ y) = -\text{RU}(-(x \circ y))$. Square roots are corrected to the required direction by an exact fused-multiply-add sign test.
- No library transcendental function enters any bound. Powers q^{-a} are formed by repeated division, $(a+1)^\theta$ by repeated multiplication and a directed square root, and the logarithms in the test of [Proposition 5.4](#) are replaced by rigorous lower bounds from a truncated artanh series with positive terms. Library functions such as `log`, `pow` and `sqrt` are called only to evaluate the schedule and to choose free parameters (enumeration cut-offs, the split point y of [Section 5.4](#), table sizes).
- Quantities that enter a total with a plus sign are rounded up; quantities that are subtracted, such as the partial Euler sums in [\(4.1\)](#), the thresholds β in [Section 5.4](#) and the enumerated mass in [\(5.5\)](#), are rounded down. Differences that would cancel badly are replaced by algebraically identical sums of nonnegative terms, such as the lost-mass recursion [\(5.3\)](#) and the suffix sums [\(5.4\)](#).
- The decimal output is formatted from integers, so the printed totals are themselves upper bounds.
- A start-up self-test aborts if directed rounding or the negation idiom does not behave as expected, or if constant folding in round-to-nearest is detected. The sources forbid `-ffast-math` and turn off floating-point contraction.

The schedule and the tilts. The value d_p of the schedule is evaluated once per prime in round-to-nearest (with `libm`'s logarithm) and stored. The tilt used everywhere afterwards, in the random numbers $s_{p'}$ for $p' > p$ and in T_X , is the double $\hat{\nu}_p = \text{RU}(1/\text{RD}(1 - d_p)) \geq 1/(1 - d_p)$. We therefore apply [Theorem 4.9](#) and [Proposition 5.4](#) with the schedule $\delta_p := 1 - 1/\hat{\nu}_p \geq d_p$, for which $\nu_p = \hat{\nu}_p$ exactly. The bounds at level p use $t \leq d_p \leq \delta_p$ in (i) and $d = \min(d_p, \frac{1}{2}) \leq \delta_p$ in (ii) and (iii), so they are admissible. All δ_p are at most $0.46 + 10^{-15} < \frac{1}{2}$. The programs assert $\hat{\nu}_p \leq p$ at every prime, and $t \leq d_p$ holds by construction (`rigx.c` uses $t = d_p$) or is asserted (`rigcert.c`).

A rounding gap and its repair. An audit found one rounding-direction flaw in `rigcert.c`: $(p-1)^2$ was formed by converting an exact integer to a double, which can round *up* in round-upward mode for $p \geq 94,906,267$, where $(p-1)^2 > 2^{53}$, although it is used as a divisor of upper bounds (in the second moments and in the factors of T_X). These bounds could therefore be low by one unit in the last place. The program now uses $\text{RD}((p-1)^2)$ and reproduces the certified output for [Theorem 1.3](#) bit for bit (`src/CERT_NOTES.md`); `rigx.c` had the repair from the start.

6.2. The certificate for [Theorem 1.1](#). The certified run for $m = 14,501$ (command in [Appendix D](#)) prints

$$\eta_{\text{up}} = 0.999926855181 \quad (A = 0.234899856304, B = 0.763569005991, C = 0.001457990070),$$

where A , B and C are the contributions of the primes $p < 50$, $50 \leq p \leq 2 \cdot 10^5$ and $2 \cdot 10^5 < p \leq 10^9$. The total is the double 0.99992685518076208 (`0x1.fff669aac979p-1`), rounded up in the twelfth decimal. The program also prints $k = \pi(10^9) = 50,847,534$, the upper bound $T_{\text{up}} = 705974.99523353134$ for T_{10^9} , and the quantities of the test of [Proposition 5.4](#). [Table 2](#) shows the budget by prime ranges, next to that of the certificate for [Theorem 1.3](#), and [Appendix B](#) gives finer breakdowns. About 91% of the budget is spent at the primes below 2,000.

TABLE 2. Where the budget goes. Certified contributions of the prime ranges for $m = 14,501$ ($P_{\max} = 10^9$) and $m = 16,000$ ($P_{\max} = 2 \cdot 10^8$); the second column gives the number of primes (for the last two rows, for $m = 14,501$ and for $m = 16,000$). Every entry is rounded up in the sixth decimal; the totals are the certified values of η_{up} , rounded up.

primes	number	$m = 14,501$	$m = 16,000$
$p < 17$	6	0.037108	0.034611
$17 \leq p < 50$	9	0.197793	0.195961
$50 \leq p < 200$	31	0.364896	0.365330
$200 \leq p < 2,000$	257	0.314659	0.312354
$2,000 \leq p < 20,000$	1,959	0.072697	0.071735
$20,000 \leq p \leq 2 \cdot 10^5$	15,722	0.011319	0.010886
$2 \cdot 10^5 < p \leq P_{\max}$	50,829,550 / 11,060,953	0.001458	0.006511
total, $p \leq P_{\max}$	50,847,534 / 11,078,937	0.999927	0.997385

Proof of Theorem 1.1. Apply Proposition 5.4 with $m = 14,501$, $X = 10^9$, the schedule $\delta_p = 1 - 1/\hat{\nu}_p$ of Section 6.1 for $p \leq X$ and $\delta_p = \frac{1}{2}$ for $p > X$, and c_p the per-prime costs computed by the certified program. Each c_p is an upper bound for one of the quantities (i)–(iii) of Theorem 4.9: for $p < 17$ the first moment, which is (i) with $t = 0 = \delta_p$, and otherwise a comparison bound with $t = d_p \leq \delta_p$, multiplied by a margin factor larger than 1. The program certifies $\eta = \sum_{p \leq X} c_p \leq 0.999926855181 < 1$, and with $\mu_{\text{lo}} = \text{RD}(1 - \eta_{\text{up}}) \geq 7.3144 \cdot 10^{-5}$ it certifies

$$\frac{T_X}{1 - \eta} \leq f_{\text{up}} = \text{RU}\left(\frac{T_{\text{up}}}{\mu_{\text{lo}}}\right) \leq 9.6518 \cdot 10^9, \quad (\log k + \log \log k - 3)^2 k \geq 1.5787 \cdot 10^{10},$$

where $k = 50,847,534 \geq 10$ and the logarithms are bounded below as in Appendix C. Since $9.6518 \cdot 10^9 < 1.5787 \cdot 10^{10}$, Proposition 5.4 applies. Therefore no family of congruences with distinct moduli, all at least 14,501, covers $\mathbb{Z}$. Corollary 1.2 follows, since a covering system with distinct moduli must then have a modulus below 14,501. $\square$

The test of Proposition 5.4 passes with $f/(\log k + \log \log k - 3)^2 k \approx 0.611$; it would hold for any η up to about 0.999955 (certificate-14501/xcheck/results/tail.out), so the margin of the certificate is about $2.8 \cdot 10^{-5}$ in η . The margin of Theorem 4.9 is not enough here, as noted in Section 5.9.

6.3. The formal certificate for Theorem 1.1. The formal proof of Theorem 1.1 (Section 6.8) uses a second checker, written in Lean (Checker2Impl). It evaluates the per-prime bounds as in Sections 5.4 and 5.5, for $m = 14,501$, the primes $p \leq X = 2 \cdot 10^9$ and $t = \delta_p$, in fixed-point arithmetic with values $n/2^{62}$ and explicit directed rounding. The schedule is that of the C certificate with the two knots at $p \geq 5 \cdot 10^4$ set to 0.45, rounded to multiples of 10^{-9} , with $\delta_p = 0$ for $p < 17$ and $\delta_p = \frac{1}{2}$ for $p > X$.

- For $p < 17$ it bounds $\mathbb{E}[U_p(s)]$, which is (i) with $t = \delta_p = 0$.
- For $17 \leq p < m/2$ it uses Lemma 5.1 in the lower-tail form $\mathbb{E}[(\tau_S T - \beta)^+] = \tau_S \mathbb{E}[T] - \mathbb{E}[\beta] + \mathbb{E}[(\beta - \tau_S T)^+]$. The last term involves the joint law of (T, b) only for $T \leq \beta/\tau_S$, so this evaluation has neither lost mass nor exponent truncation. The enumeration of s_S is pruned as in Section 5.4, at the relative level 2^{-33} , and the pruning bounds are added.

- For $m/2 \leq p \leq 2 \cdot 10^5$ it uses the τ -split (Lemma 5.2) with the truncations of Section 5.6, with the law of τ_S kept on $[0, 2^{22}]$.
- For $2 \cdot 10^5 < p \leq X$ it takes the primes in blocks of relative width 2^{-13} and evaluates the τ -split bound once per block, at the first element of the block, where it is largest.

The run certifies

$$\sum_{p \leq X} c_p \leq 0.999941815948, \quad T_X \leq 796860.64557339;$$

the primes $p < 17$, $17 \leq p \leq 2 \cdot 10^5$ and $2 \cdot 10^5 < p \leq X$ contribute 0.037107790428, 0.961380248462 and 0.001453777059.

Formal proof of Theorem 1.1. Apply Theorem 4.9 in the form of Remark 4.10, with $m = 14,501$, $X = 2 \cdot 10^9$, the schedule above, the per-prime costs c_p of the Lean checker (each an upper bound for the quantity (i) with $t = \delta_p$), and the constant $C_X = \frac{5941}{50000}$ of Lemma 4.8. The left-hand side of the criterion is at most

$$\begin{aligned} 0.999941815948 + \frac{5941}{50000} \cdot \frac{796860.64557339}{2 \cdot 10^9} \\ \leq 0.999941815948 + 0.000047341491 \leq 0.999989157439 < 1. \end{aligned}$$

Therefore no family of congruences with distinct moduli, all at least 14,501, covers $\mathbb{Z}$. $\square$

The whole argument, including the soundness of every step of the checker and the tail estimate, has been checked by Lean's kernel. The only trusted step is the compiled evaluation of the checker (Section 6.8), which takes about ten minutes and 2.6 GB. The margin is $1.08 \cdot 10^{-5}$. For $m = 14,500$ a floating-point prototype of the same checker gives $\sum_p c_p > 1$ on its own, so 14,501 is also the limit of the formal version of the method. On the way, the same development certified $m = 14,600$ with $X = 2 \cdot 10^8$ and the tail of Lemma 4.7 (total 0.999622630), and $m = 14,505$ with $X = 10^9$ (total 0.99997629). The C program `certificate-14501/src/rigx.c`, run at $m = 14,600$ and $X = 2 \cdot 10^8$ with matching settings, agrees with the Lean checker's sum $\sum_{p \leq X} c_p$ to within 10^{-6} .

6.4. The certificate for Theorem 1.3. The certified run for $m = 16,000$ prints

$$\eta_{\text{up}} = 0.997384393974 \quad (A = 0.230571210247, B = 0.760302397508, C = 0.006510785610),$$

for the primes $p < 50$, $50 \leq p \leq 2 \cdot 10^5$ and $2 \cdot 10^5 < p \leq 2 \cdot 10^8$. The total is the double 0.9973843939732947 (0x1.fea92ad34fe9cp-1), rounded up in the twelfth decimal. The program also prints $k = \pi(2 \cdot 10^8) = 11,078,937$ and the upper bound $T_{\text{up}} = 321051.36453405279$ for $T_{2 \cdot 10^8}$.

Proof of Theorem 1.3. Apply Theorem 4.9 with $m = 16,000$, $X = 2 \cdot 10^8 \geq 2^{27}$, the schedule $\delta_p = 1 - 1/\hat{\nu}_p$ of Section 6.1 for $p \leq X$ and $\delta_p = \frac{1}{2}$ for $p > X$, and c_p the per-prime costs computed by the certified program, each of which is an upper bound for one of the quantities (i)–(iii). The program certifies $\sum_{p \leq X} c_p \leq 0.997384393974$ and $T_X \leq T_{\text{up}} < 321051.3646$. Hence the left-hand side of (4.3) is at most

$$0.997384393974 + \frac{100}{189} \cdot \frac{321051.3646}{2 \cdot 10^8} \leq 0.997384393974 + 0.000849343 < 0.998234 < 1.$$

Therefore no family of congruences with distinct moduli, all at least 16,000, covers $\mathbb{Z}$. $\square$

The criterion of BBMST gives an independent confirmation of the tail. With $\mu_{10} = \text{RD}(1 - \eta_{\text{up}}) \approx 0.0026156$, the program certifies $T_X/(1 - \eta) \leq f_{\text{up}} \leq 1.2275 \cdot 10^8$ and $(\log k + \log \log k -$

$3)^2k \geq 2.8386 \cdot 10^9$, so [Proposition 5.4](#) applies as well, with $f/(\log k + \log \log k - 3)^2k \approx 0.043$ (`logs/xcheck_final.md`).

6.5. Independent recomputation. *The certificate for [Theorem 1.1](#).* Two independent checks were made.

- (1) *Cross-check by independently written code* (`certificate-14501/xcheck/`). The programs of the recomputation of the certificate for [Theorem 1.3](#) described below, which were written independently of both checkers, were rerun with the new schedule (only their schedule header was changed), together with a new program for the large primes that uses no code of `rigx.c`. For each prime they compute a rigorous interval $[lo, hi]$ for the exact value of the level bound. [Table 3](#) summarizes the results: the certified cost is at least hi at every prime compared, and the total of the upper ends for $2 \cdot 10^5 < p \leq 10^9$ is below the certified C .
- (2) *Independent referee.* An AI agent acting as a referee checked the mathematics of [Sections 5.4 to 5.6](#) by hand; brute-forced the identity of [Lemma 5.1](#) with exact rationals on 36,840 random pairs (p, s) at all 921 primes where it is used, including cases with y itself a prime of B and with $v_q \geq 2$ for $q \in B$ (no mismatch); audited the rounding directions and the lost-mass accounting; and recomputed every level bound for all primes up to 10^9 with separately written programs (an exact complement identity with a pruned enumeration for $p < 7,253$, and a logarithmic-grid law of $\tau(s)$ with its own sieve for the larger primes), with exact-rational and 60-digit spot checks. It obtained the rigorous interval $[0.99992148081, 0.99992282962]$ for the sum of the level bounds, found the certified cost at or above its upper bound at every prime (at every level up to $2 \cdot 10^5$ and at every 500-th prime beyond, where the per-prime costs are recorded), reproduced the run bit for bit on macOS with Apple clang and with GNU gcc, and found no errors. Its remaining remarks were that the trust base includes [\[1, Theorem 6.1\]](#) and that the bound for the largest primes is conservative ([Section 5.5](#)).

TABLE 3. The cross-check of the certificate for [Theorem 1.1](#). Here hi is the upper end of an independent rigorous interval for the exact level bound, and “cost” is the certified cost, including the margins of [Section 5.1](#).

check	result
schedule, all 17,984 primes $p \leq 2 \cdot 10^5$	relative deviation from a 40-digit evaluation at most $4.5 \cdot 10^{-16}$; $\hat{\nu}_p \geq 1/(1 - d_p)$ everywhere
first moments, $p < 17$	exact rational values; at most the certified cost at all 6 primes
S/B/H levels, $17 \leq p \leq 7,247$	all 921 primes, with two different splits whose intervals always overlap; 0 violations; $cost/hi \in [1 + 9.2 \cdot 10^{-7}, 1 + 9.6 \cdot 10^{-7}]$
τ -split levels, $7,253 \leq p \leq 2 \cdot 10^5$	all 17,057 primes, with a direct law of $\tau(s)$; 0 violations; $cost/hi \in [1 + 4.5 \cdot 10^{-5}, 1 + 1.2 \cdot 10^{-4}]$
τ -split levels, $2 \cdot 10^5 < p \leq 10^9$	118,717 primes compared (all $p \leq 2 \cdot 10^5$ and every 500-th prime beyond), 0 violations; the upper ends summed over all 50,829,550 primes give $0.0014578122 \leq C$
tail	$k = 50,847,534$ confirmed by a segmented sieve; $T_k = 705974.98819$, below T_{up} by $1.0 \cdot 10^{-8}$ relative; $f/(\log k + \log \log k - 3)^2k = 0.611$

The certificate for [Theorem 1.3](#). Separately written code (`tests/xcheck_final/`), with its own decomposition and its own outward rounding, recomputed this certificate (`logs/xcheck_`

`final.md`); it reuses none of the checker’s machinery. For each of the 17,984 primes $p \leq 2 \cdot 10^5$ and each candidate bound of the checker, it computed a rigorous interval $[\text{lo}, \text{hi}]$ for the exact value and checked $c_p \geq \min_{\text{candidates}} \text{hi}$ in exact rational arithmetic: there were 0 violations and 0 inconclusive cases, and the smallest relative margin, $1.4 \cdot 10^{-7}$ at $p = 8009$, is seven times the width of the interval there. For $31 \leq p < 8000$ it uses the identity of [Lemma 5.1](#), and for $8000 < p \leq 2 \cdot 10^5$ a divisor-count dynamic program on $[1, 2^{20}]$ and the exact identity $\mathbb{E}[(\tau - x)^+] = \mathbb{E}\tau - x + \sum_{T \leq x} (x - T) \mathbb{P}(\tau = T)$, which needs no truncation; both are run in both rounding directions and agree with brute force on small instances. It also reproduced the schedule bit for bit, the first and second moments (against 60-digit values and direct double sums), the total 0.0065107856047 of the fractional-moment bounds over $(2 \cdot 10^5, 2 \cdot 10^8]$ (against the checker’s 0.006510785610), the prime count, the product $T_k = 321051.3638366$ (the checker’s T_{up} is larger by $2.2 \cdot 10^{-9}$ relative) and the tail criterion. It also measures the slack of the cruder evaluation. The sum over $p \leq 2 \cdot 10^5$ of the exact values of the best candidate bounds is 0.98120, against the checker’s 0.990874; the difference, about 0.0097, is almost entirely at $p \geq 50$. The recomputations check numbers, not mathematics: the comparison theorem and the reduction of [Section 4](#) are outside their scope. Their interval code relies on the compiler honouring directed rounding (it passes its own self-tests and was cross-validated against brute force and 60-digit arithmetic).

6.6. Tests. The checkers’ formulas. On small instances, independent Python programs compare every quantity that `rigcert.c` computes with brute-force enumeration of s (with a rigorous bracket for the pruned part), with exact rational arithmetic (for $\mathbb{E}[U_p]$ and $\mathbb{E}[U_p^2]$, via a different decomposition) and with 50-digit arithmetic (for the products $\mathbb{E}[\tau(s)^{\theta}]$) (`tests/cert/run_small.sh`). The instances use the knot schedule and the constant band of the final run, values $\delta > \frac{1}{2}$, and tiny truncation parameters that force every tail path. There were no failures over 308 comparison-bound candidates, 84 moment-bound candidates and the first and second moments at every prime; on two larger instances, 108 primes of exact moments and 828 pairs (p, θ) of 50-digit products agreed. Earlier validations on other configurations, including 431 exact prime checks, are described in `src/CERT_NOTES.md`. For `rigx.c`, plain enumeration of s was compared with 42 cases at $m = 300$ and $m = 1000$, including settings that force every truncation and lost-mass path (`certificate-14501/validation/`); the certified values were always at least the lower end of the brute-force bracket, and inside the bracket with the default settings. At $m = 16,000$ with the schedule of [Theorem 1.3](#), `rigx.c` agrees with the exact intervals of the recomputation above to within 10^{-9} relative at the S/B/H levels and lies inside the intervals at the others.

The comparison theorem. About 10^6 randomized, exhaustive and adversarial tests (`tests/review/`) found no violation of [Lemma 3.4](#), [Theorem 3.2](#) and [Lemma 4.1](#). They compare the theorem with the exact worst case over all ν -bounded product measures (computed by a backward bathtub recursion), include an exhaustive search over all families of at most three cylinders on $\mathbb{Z}/4 \times \mathbb{Z}/3$, and simulate the measures of BBMST exactly on small moduli, checking [Lemma 2.1](#), the kernel bounds of [Lemma 2.7](#), [Lemma 2.3](#) and each inequality in the proof of [Theorem 4.4](#).

Proof review. Separate AI referee passes checked each step of the proof of the comparison theorem and of the reduction, and the new steps of [Sections 5.4](#) to [5.6](#); they found no gap, and their presentational suggestions have been incorporated. No human has refereed the argument yet.

6.7. Run times and reproducibility. The certified run for [Theorem 1.1](#) takes 87–93 seconds on one core of an Apple M3 Max and about 1.2 GB of memory (for the sieve up to 10^9). Rebuilt

from the published source, it reproduces the certified result lines and the per-prime dump bit for bit; the referee reproduced them on macOS with Apple clang and with GNU gcc. On Linux (x86-64, glibc, GNU gcc at `-O0` and at `-O2`, which agree) the run also certifies, with the same result line; the last digits of the certified value and 503 of the 119,644 per-prime costs in the dump differ, by at most about $2 \cdot 10^{-12}$ relative, because of the logarithm (see below). The cross-check of [Table 3](#) takes about 29 minutes on one core. The certified run for [Theorem 1.3](#) takes 156.6 seconds of CPU time on one core (`logs/timing.txt`) and is also reproduced bit for bit (`logs/repro16k.txt`). Bit-for-bit reproduction requires floating-point contraction to be off and the same `libm` logarithm, which is used only to evaluate the schedule; another `libm` may move some d_p by one unit in the last place. The last digits of the output would then change, but the certificates would remain valid, since any schedule is admissible and the checkers use their own values consistently.

6.8. Formal verification. [Theorems 1.1](#) and [1.3](#) have been formally verified in Lean 4 (v4.34.1) with Mathlib, each modulo one `native_decide` evaluation. The development accompanies this paper, and its `README.md` describes the build and the audit. The root theorems, in `MinModulus/Checker2Sound/Final14501.lean` and `MinModulus/Main/Main.lean`, are

```
theorem MinModulus.Checker2Sound.not_covers_14501 {ι : Type*} [Fintype ι]
  (d : ι → ℕ) (a : ι → ℤ) (hd : Function.Injective d)
  (hm : ∀ i, 14501 ≤ d i) :
  ∃ x : ℤ, ∀ i, ¬ ((d i : ℤ) | x - a i)
theorem MinModulus.not_covers {ι : Type*} [Fintype ι]
  (d : ι → ℕ) (a : ι → ℤ) (hd : Function.Injective d)
  (hm : ∀ i, 16000 ≤ d i) :
  ∃ x : ℤ, ∀ i, ¬ ((d i : ℤ) | x - a i)
```

Neither contains `sorry`. For each, `#print axioms` lists only `propext`, `Classical.choice`, `Quot.sound` and one auxiliary axiom introduced by `native_decide`:

```
MinModulus.Tail2.check2T14501_eq_true._native.native_decide.ax_1_1
MinModulus.CheckerImpl.check_eq_true._native.native_decide.ax_1_1
```

respectively. Each states that one closed Boolean (`check2T14501`, respectively `check`) evaluates to `true`. The trust base is therefore Lean’s kernel together with the compiled evaluation of this Boolean. That takes about ten minutes and 2.6 GB for `check2T14501`, and about 50 seconds for `check`. It trusts Lean’s compiler and runtime, including the arbitrary-precision arithmetic on natural numbers. The code of both checkers contains no `implemented_by`, `extern`, `unsafe` or `partial` declarations; the only compiler rewrites are two `@[csimp]` lemmas, which are proved equalities. The evaluation of `check` has been re-run from a fresh file. The whole development, including every native evaluation, has also been rebuilt from a fresh copy of the sources (with the Mathlib build shared), in about eleven minutes. Both gave the same results and the same axioms.

The formalization of [Theorem 1.1](#). It reuses the modules `Arith`, `Distortion`, `Comparison`, `Smooth`, `Tail` and `Main` of the first formalization unchanged, including the certificate interface `Cert`. It also reuses the checker’s arithmetic, sieve and first-moment code with their soundness proofs. New are the following parts.

- The checker `Checker2Impl`, which implements [Section 6.3](#).
- The mathematics of its quantities, in `Checker2Math`:
 - [Lemmas 5.1](#) and [5.2](#) and the domination $\tau(s) \leq 2 \sum v_q$;
 - the lower-tail identity;
 - the joint law of (T, b) and the law of the exponent sum;

- the depth-first enumeration and its pruning bound.
- A code-level soundness proof, `Checker2Sound`, covering:
 - the dynamic programs, their states and the tables;
 - the S/B/H and τ -split costs;
 - the loops and the blocks;
 - the assembly of the certificate interface (`E.run2_sound`).
- The sharpened tail estimate [Lemma 4.8](#) and the corresponding form of [Theorem 4.9](#) (`Tail2`), and the certificate `check2T14501_eq_true`.

The soundness theorem `E.run2_sound` holds for all parameters of the checker that satisfy four decidable side conditions. A certificate for another m , or another schedule, therefore needs only a new `native_decide` evaluation.

Structure of the first formalization. The whole development has 95 files and about 28,000 lines, counting the formalizations of [Theorems 1.1](#) and [1.3](#). Of these, 50 files and about 16,400 lines belong to the first formalization. [Table 4](#) lists the modules and their correspondence with this paper. The first formalization follows the paper, with two differences: it uses the comparison theorem only for the hinge functions, since the other bounds are obtained from the hinge with $t = \delta_p$ by pointwise inequalities on the aligned side, and it treats the primes above $X = 2 \cdot 10^8$ by [Lemma 4.7](#), so that it needs neither [\[1, Theorem 6.1\]](#) nor Dusart’s estimate.

The checker of the first formalization. The numerical certificate for [Theorem 1.3](#) is produced by an independent implementation of the checker in core Lean (`CheckerImpl`), not a transcription of `src/rigcert.c`. It works in fixed point, with values $n/2^{62}$ and explicit directed rounding; it uses $t = \delta_p$ only, with δ_p an exact rational (a multiple of 10^{-9} obtained from the same knots) and exact rational tilts; and for $2 \cdot 10^5 < p \leq 2 \cdot 10^8$ it processes the primes in blocks, taking the factors of each block at its first prime, with $\theta \in \{2, \frac{5}{2}, 3\}$. It certifies $\sum_{p \leq 2 \cdot 10^8} c_p \leq 0.997384786285$ and $T \leq 321095.49941797$ (slightly larger than in the C program, because of the blocks), so that the total with the tail term $\frac{100}{189}T/X = 0.000849458994$ is $0.998234245279 < 1$. A code-level soundness proof (`CheckerSound`) shows that the checker’s loops and arrays bound the true per-prime losses (invariants of the dynamic programs, semantics of the enumeration, the blocks) and assembles them into the certificate `Cert 16000 (2*10^8)  $\delta$  c T` consumed by `MinModulus.Main.not_covers_of_cert`, whose conditions correspond to those of [Theorem 4.9](#) with $t = \delta_p$.

What is not formalized. The C certificate of [Theorem 1.1](#) ([Section 6.2](#)) and [\[1, Theorem 6.1\]](#), which it uses, are not formalized; the formal proof does not need them.

7. LIMITS OF THE METHOD

The proof bounds the true loss $P_i(B_i)$ by a chain of steps: the union bound $\alpha_i \leq \min(1, U_i)$ in the fibre ([Lemma 2.8](#)); the pointwise hinge step, which discards the cap $\alpha_i \leq 1$ ([Lemma 4.1](#)); the comparison theorem, which aligns the progressions and tilts the measure ([Theorem 3.2](#)); the enlargement to all admissible moduli ([Lemma 4.3](#)); and the union bound over the levels in [Proposition 2.4](#). On top of this comes the slack of the numerical evaluation. In this section we discuss how much each step loses. Statements labelled Proposition are proved. Statements labelled Observation are numerical findings from floating-point experiments (Monte Carlo sampling at $m = 16,000$, and exact computations and adversarial searches on small analogues, and computations of the exact level bounds near $m = 14,500$); they are not proofs and are not used anywhere in the proofs of [Theorems 1.1](#) and [1.3](#).

TABLE 4. Modules of the Lean formalization (MinModulus/) and the corresponding results of this paper.

module	lines	contents
Arith	679	setup of a family of congruences: Q , the primes and levels, the Chinese remainder theorem, the sets B_ℓ , Lemmas 2.8, 4.1 and 4.3
Distortion	646	the measures (2.3), kernels and tilts, Lemmas 2.1 to 2.3 , Proposition 2.4 and Lemma 2.7
Comparison	710	Lemmas 3.3 and 3.4 (by duality), Proposition 3.6 and Lemma 3.7 , and Theorem 3.2 for hinges
Smooth	2057	the tilted law of s , $U_p(s)$, the moments $\mathbb{E}[\tau(s)^\theta]$, hinge bounds, monotone coupling in the exponent caps
Tail	354	Lemma 4.7
Main	594	the certificate interface <code>Cert</code> , <code>not_covers_of_cert</code> (Theorem 4.4 with $t = \delta_p$, Theorem 4.9), the final theorem
CheckerImpl	1201	the executable checker and <code>check_eq_true</code> (by <code>native_decide</code>)
CheckerMath	4011	the mathematics of the checker's quantities: laws of $\tau(s_A)$ and $\tau(s_B)$, the split of Lemma 5.3 , the comparison bound, first and fractional moments
CheckerSound	6130	code-level soundness: loop and array invariants, enumeration, dynamic programs, blocks, and the assembly of the certificate (<code>CheckerSound.D.cert_16000</code>)
<i>formalization of Theorem 1.1</i>		
Checker2Impl	1073	the executable checker of Section 6.3
Checker2Math	2064	Lemmas 5.1 and 5.2 in the form of the hinge loss, the lower-tail identity, the joint law of (T, b) , the law of the exponent sum, the enumeration and its pruning bound
Checker2Sound	6512	code-level soundness: dynamic programs, tables, the S/B/H and τ -split costs, loops and blocks, the generic soundness theorem <code>E.run2_sound</code> , and the final theorem <code>not_covers_14501</code>
Tail2	1950	Lemma 4.8 , the criterion with the sharpened tail, and the certificate <code>check2T14501_eq_true</code> (by <code>native_decide</code>)

7.1. Steps that are sharp. *Alignment and tilt.* By [Proposition 3.9](#), [Theorem 3.2](#) is attained by aligned cylinders and a suitable ν -bounded measure, so no better inequality holds for the whole class of ν -bounded product measures.

Enlargement. [Lemma 4.3](#) loses nothing for a family that contains all admissible moduli $m'p^j \geq m$ with $m' \mid Q_{i-1}$, up to the truncation of the exponents, which disappears as the exponents γ_j grow. Such families can be used at all levels at once, since moduli with different largest prime factors are automatically distinct.

The redistribution rule. By [Remark 2.5](#), given P_{i-1} the rule (2.3) minimizes the loss at level i among all rules that preserve the marginal and multiply masses by at most ν_i .

The tilt in practice. [Proposition 3.9](#) uses a measure that is chosen freely, whereas P_{i-1} is produced by the rule (2.3). The following observation shows that the tilt is nevertheless essentially attained.

Observation 7.1 (the tilt is realized by nested shells). Consider configurations of *nested shells*, in which the progressions of every level are aligned at a common centre, so that points that are heavily covered at one level are heavily covered at every smaller level as well. In such a configuration the fibres of rich points receive, at each earlier level, the tilt $\min(\nu_q, 1/(1 - \alpha_q))$

of (2.3). Monte Carlo sampling (10^6 importance-sampled paths) at $m = 16,000$ with the final schedule, for $p \leq 2 \cdot 10^4$, gives the following totals of the per-level bound $\mathbb{E}[(U_p - \delta_p)^+]/(1 - \delta_p)$: 0.970 with the full tilt ν_q (the bound used in the proof, whose exact value is 0.9703), 0.767 without tilt ($\nu_q = 1$), and 0.953 with the tilt that nested shells realize. So the tilt accounts for about 0.20 of the budget, most of it at $200 \leq p < 8,000$, and all but about 0.017 of it is realized by an actual configuration.

7.2. The cap cannot be used by convex functionals. The pointwise step replaces $(\min(1, U) - \delta)^+$ by the hinge $(U - t)^+$, which grows beyond $U = 1$. The next proposition shows that no convex functional can do better, so that the comparison theorem cannot exploit the cap $\alpha \leq 1$.

Proposition 7.2. *Let $0 < \delta < 1$ and $f_\delta(u) = (\min(1, u) - \delta)^+/(1 - \delta)$ for $u \geq 0$. Let $\varphi: [0, \infty) \rightarrow \mathbb{R}$ be convex with $\varphi \geq f_\delta$. Then for every random variable $W \geq 0$ with $\mathbb{E}W < \infty$,*

$$\mathbb{E}\varphi(W) \geq \min\left(1, \min_{0 \leq t \leq \delta} \frac{\mathbb{E}(W - t)^+}{1 - t}\right).$$

Proof. Replacing φ by its lower semicontinuous regularization, which is still convex, still majorizes the continuous function f_δ and is not larger than φ , we may assume that φ is continuous. Let $\sigma^* = 1/(1 - \delta)$, the slope of f_δ on $[\delta, 1]$, and let $\varphi'_-(1)$ be the left derivative at 1.

Case $\varphi'_-(1) \leq \sigma^$.* Put $\sigma = \varphi'_-(1)$; it is a subgradient at the interior point 1, so $\varphi(u) \geq \varphi(1) + \sigma(u - 1) \geq 1 + \sigma(u - 1)$ for all $u \geq 0$, because $\varphi(1) \geq f_\delta(1) = 1$. Also $\varphi \geq 0$. If $\sigma \leq 0$, then $\varphi \geq 1$ on $[0, 1]$, and $\varphi \geq f_\delta = 1$ on $[1, \infty)$, so $\mathbb{E}\varphi(W) \geq 1$. If $0 < \sigma \leq \sigma^*$, put $t = 1 - 1/\sigma \leq \delta$; then $\varphi(u) \geq \max(0, \sigma(u - t)) = \max(0, (u - t)/(1 - t))$. For $t \geq 0$ this gives $\mathbb{E}\varphi(W) \geq \mathbb{E}(W - t)^+/(1 - t)$. For $t < 0$ it gives $\mathbb{E}\varphi(W) \geq (\mathbb{E}W - t)/(1 - t)$, a convex combination of $\mathbb{E}W$ and 1, which is at least $\min(1, \mathbb{E}W)$, the bound for $t = 0$.

Case $\varphi'_-(1) > \sigma^$.* The continuous convex function $\psi(u) = \varphi(u) - \sigma^*u$ attains its minimum over $[0, 1]$ at some $u^* < 1$, because $\psi'_-(1) > 0$. It is then a global minimum on $[0, \infty)$: if $\psi(v) < \psi(u^*)$ for some $v > 1$, convexity would give $\psi(w) < \psi(u^*)$ for $w \in (u^*, 1]$. Hence $\varphi(u) \geq \varphi(u^*) + \sigma^*(u - u^*)$ for all $u \geq 0$. Since $\varphi(u^*) \geq f_\delta(u^*) \geq (u^* - \delta)/(1 - \delta)$, this gives $\varphi(u) \geq (u - \delta)/(1 - \delta)$, and together with $\varphi \geq 0$, $\varphi(u) \geq (u - \delta)^+/(1 - \delta)$. So $\mathbb{E}\varphi(W) \geq \mathbb{E}(W - \delta)^+/(1 - \delta)$. $\square$

Applied to $W = U^{\text{al}}$, Proposition 7.2 says that the bound of Theorem 4.4, minimized over t and capped at 1, is the best that Theorem 3.2 can give for the loss. The cap does matter at the aligned configuration: there the excess $\mathbb{E}[(U - 1)^+]/(1 - \delta)$ charged beyond the cap amounts to 0.217 (numerical), about 22% of the budget for $p \leq 2 \cdot 10^4$, and between 20% and 26% of the budget of each range of primes. An adversary, however, can spread the progressions apart.

Observation 7.3 (spreading). In exact computations on small analogues (moduli built from the primes up to 7 or up to 13, with the worst ν -bounded measure and adversarial residues found by simulated annealing), spreading the progressions recovers most of the gap between the capped aligned value and the per-level bound: between 59% and 100%, depending on the instance. At $m = 16,000$, the explicit spreadings we tried recover at most about 15% of the cap excess at the prime 101. Whether the true per-level worst case at this scale is much below the bound of Theorem 4.4 is open.

7.3. Cross-level accounting. The union bound over the levels in Proposition 2.4 is not an identity, and its slack has a simple exact description.

Proposition 7.4. *Call $x \in \mathbb{Z}_{Q_{i-1}}$ alive if $x \notin \bigcup_{j < i} B_j$. Then*

$$P_n\left(\bigcup_{i=1}^n B_i\right) = \sum_{i=1}^n \frac{\mathbb{E}_{P_{i-1}}[(\alpha_i - \delta_i)^+ \mathbf{1}\{x \text{ alive}\}]}{1 - \delta_i}.$$

Proof. The union is the disjoint union of the sets $B_i \setminus \bigcup_{j < i} B_j$, which are Q_i -measurable, so by [Lemma 2.1](#) its P_n -measure is $\sum_i P_i(B_i \setminus \bigcup_{j < i} B_j)$. The set $\bigcup_{j < i} B_j$ is Q_{i-1} -measurable, so each fibre $F(x)$ either lies inside it or is disjoint from it. Hence $P_i(B_i \setminus \bigcup_{j < i} B_j) = \sum_{x \text{ alive}} P_i(F(x) \cap B_i)$, and the computation in the proof of [Lemma 2.3](#) finishes the proof. $\square$

Thus $\sum_i P_i(B_i)$ also charges the losses on fibres that are already covered at an earlier level. In the nested shells of [Observation 7.1](#) this happens systematically: the same rich points are charged at every level.

Observation 7.5 (double counting). On two small exact analogues (moduli dividing $2^4 3^2 5 \cdot 7$ and $2^5 3^3 5 \cdot 7$, with schedules optimized for the bound of [Theorem 4.4](#)), the per-level bounds sum to 1.0241 and 0.9938; simulated annealing finds systems with $\sum_i P_i(B_i) = 0.9934$ and 0.9511, so the per-level chain loses only 0.031 and 0.043. Simulated annealing aimed directly at $P_n(\bigcup_i B_i)$ finds at most 0.907 and 0.886: the double counting, about 0.07 to 0.09 of the budget, exceeds the whole per-level slack. At $m = 16,000$, in a Monte Carlo model of nested shells, the configuration that attains the per-level bounds simultaneously (sum of capped losses about 0.74 for $p \leq 2 \cdot 10^4$) newly covers only 0.10–0.16 of the mass. We did not obtain a rigorous improvement of the certified total from this observation.

7.4. The slack of the evaluation and the reach of the method. For the certificate of [Theorem 1.3](#) the evaluation itself is lossy. At $m = 16,000$, with the schedule of that certificate, the exact values of the level bounds (computed with `rigx.c`, without margins) total 0.982431 over the primes up to $2 \cdot 10^8$, against the certified 0.997384393974. About 0.0097 of the difference comes from $50 \leq p < 200$, where the decomposition of [Lemma 5.3](#) is inexact once $N_1 > 53$, and about 0.0053 from $p > 2 \cdot 10^5$, where the fractional-moment bounds are about six times the exact values. The exact evaluation of [Sections 5.4 to 5.6](#) removes this slack. The following observation, from computations of the exact level bounds without the safety margins and with heuristically optimized schedules, describes the reach of the per-level method.

Observation 7.6 (the frontier). With the exact evaluation and the schedule of [Theorem 1.3](#), the total crosses 1 near $m = 14,780$. With the 103-knot schedule of [Theorem 1.1](#) and $P_{\max} = 10^9$, the total is 0.9999918 at $m = 14,500$ and 1.0000041 at $m = 14,499$. One damped per-prime Newton step on the schedule (a schedule with 3,241 knots) gives 0.99996996 at $m = 14,498$ and 1.00002227 at $m = 14,495$. A total below 1 is not yet a proof: at $P_{\max} = 10^9$ the test of [Proposition 5.4](#) needs $1 - \eta \geq T_X / (\log k + \log \log k - 3)^2 k \approx 4.5 \cdot 10^{-5}$, which is why $m = 14,498$ to 14,500 are not certified. We estimate the frontier of the per-level method at $m^* \approx 14,496 \pm 3$. Near $m = 14,500$ the total is a step function of m that increases on average by about 0.015 for each decrease of 1,000 in m , and about 91% of the budget is spent at the primes below 2,000, where the level bound is now evaluated exactly.

A sharper evaluation of the same level bounds can therefore gain only a few units in m . Going substantially further requires changing the method itself: a different rule for redistributing the removed mass, an accounting across levels that exploits [Proposition 7.4](#) (for instance a comparison theorem for the sub-stochastic kernels $K_j \mathbf{1}\{\text{alive}\}$), or a non-convex treatment of the cap.

Acknowledgements. This work was carried out with substantial assistance from AI systems (Anthropic’s Claude), including the design of the computation, the proofs, independent re-verification and the Lean formalization; all computations are reproducible from the accompanying code. The results have been checked by independent recomputation, by randomized, exhaustive and brute-force tests and by separate AI referee passes over the proofs and the computations. [Theorems 1.1](#) and [1.3](#) have been formally verified in Lean 4 modulo one `native_decide` evaluation each. None of the results has yet been refereed by human experts. We thank the authors of [1], whose distortion method and careful exposition this work builds on.

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APPENDIX A. THE DISTORTION SCHEDULES

Both certified programs evaluate the schedule in the same way, in double precision with round-to-nearest. Given knots $(q_0, \delta_0), \dots, (q_n, \delta_n)$ and a threshold P_D , the value at a prime

$p \geq P_D$ is $d_p = \delta_0$ if $\log p \leq \log q_0$, $d_p = \delta_n$ if $\log p \geq \log q_n$, and otherwise

$$d_p = (1 - u) \delta_l + u \delta_{l+1}, \quad u = \frac{\log p - \log q_l}{\log q_{l+1} - \log q_l},$$

where l is the first index with $\log p \leq \log q_{l+1}$. For $m = 14,501$, $P_D = 17$ and $d_p = 0$ for $p < 17$; the 103 knots are listed in [Table 5](#) (and in `certificate-14501/schedule/sched_14501.json`). For $m = 16,000$, $P_D = 50$, $d_p = 0.05$ for $31 \leq p < 50$ and $d_p = 0$ for $p < 31$; the knots are listed in [Table 6](#). The tilt is $\hat{\nu}_p = \text{RU}(1/\text{RD}(1 - d_p))$, and the distortion parameter of the proofs is $\delta_p = 1 - 1/\hat{\nu}_p$ ([Section 6.1](#)). In the programs' command lines the schedules are given by the environment variables `KNOTS`, `PD`, `PSB` and `DSB` ([Appendix D](#)).

TABLE 5. The 103 knots (q, δ) of the schedule for $m = 14,501$.

q	δ	q	δ	q	δ	q	δ	q	δ
17	0.003750	19	0.005000	23	0.006250	29	0.008750	31	0.036250
37	0.036250	41	0.055000	43	0.073750	47	0.072500	53	0.098150
59	0.106800	61	0.119050	67	0.123650	71	0.131380	73	0.141290
79	0.155230	83	0.160970	89	0.172510	97	0.178660	101	0.182540
103	0.188650	107	0.198220	109	0.204200	113	0.206020	127	0.209837
131	0.217630	137	0.223648	139	0.230595	149	0.234931	151	0.236266
157	0.238825	163	0.241289	167	0.242881	173	0.249996	179	0.259213
181	0.261402	191	0.264497	193	0.266549	197	0.269965	199	0.271361
211	0.267464	223	0.264006	227	0.266240	229	0.268576	233	0.270686
239	0.274956	241	0.279675	251	0.288000	257	0.296790	263	0.298855
269	0.299341	271	0.301179	277	0.299224	281	0.300443	283	0.299810
293	0.294209	307	0.288986	311	0.288947	313	0.290166	317	0.292580
331	0.305797	337	0.309213	347	0.314773	349	0.315866	353	0.318408
359	0.319916	367	0.318618	373	0.317664	379	0.316724	383	0.316106
389	0.315191	397	0.313993	401	0.313793	409	0.315719	419	0.313074
421	0.313538	431	0.310826	433	0.311277	439	0.310119	443	0.311003
449	0.317314	457	0.319036	461	0.319885	463	0.322807	467	0.323646
479	0.322280	487	0.337726	491	0.352946	499	0.358381	550	0.326000
650	0.348360	800	0.342500	1,000	0.365220	1,200	0.363800	1,600	0.373440
2,000	0.382450	2,700	0.387760	3,500	0.398100	5,000	0.404570	8,000	0.414350
20,000	0.434850	50,000	0.449200	200,000	0.460000				

TABLE 6. The knots of the schedule for $m = 16,000$ and the corresponding tilts $\nu = 1/(1 - \delta)$.

q	50	80	130	220	400	800	2,000	8,000	50,000
δ	0.07	0.1612	0.2191	0.2679	0.3198	0.345	0.3762	0.4031	0.4092
ν	1.0753	1.1922	1.2806	1.3659	1.4702	1.5267	1.6031	1.6753	1.6926

APPENDIX B. THE BUDGETS BY PRIME RANGES

[Table 7](#) refines [Table 2](#) for the certificate of [Theorem 1.1](#). For each range it gives the number of primes, the range of δ_p , the evaluation used ([Section 5.1](#)), the certified total $\sum c_p$ and the total of the certified first moments $\sum \mathbb{E}[U_p]$; the latter, 16.743973 over $p \leq 2 \cdot 10^5$, shows how much the distortion gains. For $p > 2 \cdot 10^5$ the program records the individual costs only at every 500-th prime; the row gives the certified total C .

TABLE 7. The certified budget for $m = 14,501$ by prime ranges (from `certificate-14501/output/dump_14501.txt.gz`). All sums are rounded up in the sixth decimal.

primes	number	δ_p	evaluation	$\sum c_p$	$\sum \mathbb{E}[U_p]$
$2 \leq p < 17$	6	0.0000	first moment	0.037108	0.037108
$17 \leq p < 31$	4	0.0038–0.0088	S/B/H	0.084588	0.090176
$31 \leq p < 50$	5	0.0363–0.0738	S/B/H	0.113205	0.147744
$50 \leq p < 100$	10	0.0982–0.1787	S/B/H	0.170212	0.318257
$100 \leq p < 200$	21	0.1825–0.2714	S/B/H	0.194684	0.613946
$200 \leq p < 500$	49	0.2640–0.3584	S/B/H	0.170063	1.053353
$500 \leq p < 1,000$	73	0.3277–0.3649	S/B/H	0.086675	1.038927
$1,000 \leq p < 2,000$	135	0.3638–0.3824	S/B/H	0.057922	1.246231
$2,000 \leq p < 7,253$	624	0.3825–0.4123	S/B/H	0.055035	2.771613
$7,253 \leq p < 20,000$	1,335	0.4123–0.4348	τ -split	0.017663	2.570146
$20,000 \leq p < 50,000$	2,871	0.4349–0.4492	τ -split	0.007209	2.588585
$50,000 \leq p \leq 2 \cdot 10^5$	12,851	0.4492–0.4600	τ -split	0.004111	4.267893
$p \leq 2 \cdot 10^5$	17,984			0.998469	16.743973
$2 \cdot 10^5 < p \leq 10^9$	50,829,550	0.4600	τ -split	0.001458	

Table 8 does the same for the certificate of Theorem 1.3, with ranges between consecutive knots of its schedule. It also gives the total $\sum b_p$ of the moment bounds of BBMST with the same schedule, and the ratio $\sum b_p / \sum c_p$. Over $p \leq 2 \cdot 10^5$ the first moments total 16.377278 and the moment bounds of BBMST 2.184877.

TABLE 8. The certified budget for $m = 16,000$ between consecutive knots of the schedule (from `logs/dump16k.txt`). All sums are rounded up in the sixth decimal.

primes	number	δ_p	$\sum c_p$	$\sum \mathbb{E}[U_p]$	$\sum b_p$	ratio
$2 \leq p < 31$	10	0.0000	0.120331	0.120331	0.120331	1.00
$31 \leq p < 50$	5	0.0500	0.110242	0.141679	0.141679	1.29
$50 \leq p < 80$	7	0.0813–0.1588	0.126588	0.214777	0.214777	1.70
$80 \leq p < 130$	9	0.1656–0.2163	0.117619	0.274202	0.270831	2.30
$130 \leq p < 220$	16	0.2198–0.2640	0.127195	0.441578	0.302992	2.38
$220 \leq p < 400$	31	0.2691–0.3191	0.123609	0.694849	0.291946	2.36
$400 \leq p < 800$	61	0.3199–0.3449	0.100274	0.978090	0.253437	2.53
$800 \leq p < 2,000$	164	0.3454–0.3762	0.082401	1.587360	0.238184	2.89
$2,000 \leq p < 8,000$	704	0.3762–0.4031	0.056629	2.956076	0.205359	3.63
$8,000 \leq p < 50,000$	4,126	0.4031–0.4092	0.022132	4.839575	0.113938	5.15
$50,000 \leq p \leq 2 \cdot 10^5$	12,851	0.4092	0.003860	4.128765	0.031409	8.14
$p \leq 2 \cdot 10^5$	17,984		0.990874	16.377278	2.184877	2.21
$2 \cdot 10^5 < p \leq 2 \cdot 10^8$	11,060,953	0.4092	0.006511			

APPENDIX C. ROUNDING CONVENTIONS OF THE CHECKERS

This appendix summarizes `src/CERT_NOTES.md` and the header of `certificate-14501/src/rigx.c`. Throughout, “up” means that the computed double is an upper bound for the exact

value of the formula applied to the computed inputs, and “down” a lower bound; the programs are in round-upward mode, and lower bounds use the negation idiom of [Section 6.1](#).

Build and self-test. The sources contain `#pragma STDC FENV_ACCESS ON` and `FP_CONTRACT OFF` and refuse to compile with `-ffast-math`. With GNU `gcc`, `-frounding-math` is required and the programs must be built with `-O0`: `gcc` ignores the `FENV_ACCESS` pragma and at `-O1` and above may reuse a result across `fesetround`; Apple `clang` 21 honours the pragma at `-O2`. At start-up the programs check that directed rounding and the negation idiom work and that no constant was folded in round-to-nearest.

First moment. By (4.1) and (5.1), $\mathbb{E}[U_p(s)] = T_1/(p-1) - U_1$ with $T_1 = \prod_{q < p} (1 + \hat{v}_q/(q-1))$ and $U_1 = (p-1)^{-1} \sum_{d < N_1} \nu(d)c(d)/d$, the sum running over $(p-1)$ -smooth d . The product T_1 is rounded up and the subtracted partial sum U_1 down.

Second moment. $\mathbb{E}[U_p(s)^2]$ is computed by a finite inclusion–exclusion formula of the form $T/(p-1)^2 - 2T_1V/(p-1) + S$, where $T = \prod_{q < p} (1 + \hat{v}_q(3q-1)/(q-1)^2)$ and S , a sum of nonnegative terms (the programs assert $\hat{v}_q \leq q$, which makes them nonnegative), are rounded up, and the subtracted quantities T_1 and V are rounded down (see `src/CERT_NOTES.md` for the formula). The divisor $(p-1)^2$ is rounded down ([Section 6.1](#)), and the bound $\mathbb{E}[U_p^2]/(4d(1-d))$ has its denominator rounded down.

The exact evaluation (rigx.c). All dynamic programs (for T_H , for (T, b) given a profile, for τ_s and for (e_2, e_3)) are positive linear maps, evaluated in round-upward mode with upper bounds for the coefficients $\mathbb{P}(v_q = a)$, so every stored probability is an upper bound; in-place updates process the cells in an order in which no updated cell is read again. The numbers $c(d) = 1 - p^{1-j_0(d)}$, $1 - \frac{1}{p}$ and $t(p-1)$ are rounded down, so the thresholds β of [Section 5.4](#) and X of [Section 5.5](#) are lower bounds, and the hinges are evaluated at these lower thresholds, which is valid since they are nonincreasing in the threshold. The suffix sums (5.4), the bounds for the pruned subtrees, the lost masses (5.3) and the margin factors are rounded up.

The simpler evaluation (rigcert.c). The probabilities $\mathbb{P}(s_A)$ are carried as intervals through the enumeration, and $U_A(s_A)$ is rounded up. The dynamic programs for the laws of $\tau(s_A)$ and $\tau(s_B)$ produce upper bounds. In $G(k, u)$, the inputs u and $k/(p-1)$ are rounded up and $c = 1 + (t-u)/a$ is rounded down, which is safe because G is monotone in each. The suffix sums for $\mathbb{E}[(\tau(s_B) - c)^+]$ are rounded up. The weights $R_k = \mathbb{P}_{\text{up}}(\tau(s_A) = k) - \mathbb{P}_{\text{lo}}(\text{enumerated}, \tau(s_A) = k)$ use the enumerated mass rounded down. Finally $1 - t$ is rounded down. The program enforces $t \leq d_p$ exactly.

Fractional moments. q^{-a} is formed by repeated division in the required direction, $(a+1)^\theta$ by repeated multiplication and a directed square root, and the series (4.2) is summed with the geometric bound for its remainder given in [Section 5.8](#). The constant c_θ is rounded up: the program uses the unconstrained maximum $\frac{d}{\theta-1}(u^*)^{-\theta}$ with u^* rounded down, or $1-d$ if the lower bound for u^* exceeds 1. The products $\mathbb{E}[\tau(s)^\theta]$ are rounded up, and $(p-1)^\theta$ and $1-d$ down.

Totals and the tail tests. The total η is the sum of the costs, rounded up. For [Proposition 5.4](#), $\mu_{\text{lo}} = \text{RD}(1 - \eta_{\text{up}})$ and $f_{\text{up}} = \text{RU}(T_{\text{up}}/\mu_{\text{lo}})$; the lower bound for $(\log k + \log \log k - 3)^2 k$ uses lower bounds for the logarithms from the truncated series $\log y = 2 \sum_{i \geq 0} z^{2i+1}/(2i+1)$, $z = (y-1)/(y+1)$, whose terms are positive (with $\log 2$ from the same series); the programs check $k \geq 10$ and that the base is positive. The logarithm lower bounds were compared with 50-digit values.

Audits. `rigcert.c` was derived from a floating-point prototype, which computed in round-to-nearest, allowed t slightly above δ_p , could round the tilt below $1/(1 - \delta_p)$, computed two truncation tails with cancellation and truncated a series without bounding its remainder; all of

this is corrected in the certified program, which also refuses a debugging option that disables the tilt. A later audit found the rounding gap described in [Section 6.1](#), which has been repaired.

APPENDIX D. REPRODUCING THE COMPUTATIONS

All paths are relative to the accompanying folder.

The certificate for [Theorem 1.1](#). In `certificate-14501/`:

```
./run_certificate.sh
```

builds `src/rigx.c` with `cc -O2 -frounding-math -ffp-contract=off (-O0 under GNU gcc)`, runs `rigx 14501 1e9` with the environment of the certified run (the 103 knots in `KNOTS`, `PD=17` `PSB=0` `DSB=0`; the accuracy settings `EPS=1e-14` `TINY=1e-30` `HEFF=1e-16` `TS=4194304` `THM=1048576` `TBUD=4194304`; the margins `MARGIN=2-20` and `MARGINL=MARGINL2=2-13`), and compares the result lines with `output/cert_14501.txt` (line breaks added):

```
m=14501 eta_up=0.999926855181 (A=0.234899856304 B=0.763569005991 C=0.001457990070)
f_up=9.65174e+09 crit_lo=1.57871e+10 SUCCESS(certified)
CERT eta_up=0.99992685518076208 (0x1.fff669aac979p-1) T_up=705974.99523353134
mu_lo=7.3144819237924708e-05 f_up=9651742974.9486885 crit_lo=15787081835.245779
k=50847534 log_k_lo=17.744342183707648 loglog_k_lo=2.8760667148961203
```

It takes about 90 seconds and 1.2 GB. The cross-check of [Table 3](#) is run by `xcheck/tools/run_xcheck.sh` with the same environment (about 29 minutes), and `validation/run_bf.py` runs the brute-force comparisons of [Section 6.6](#).

The certificate for [Theorem 1.3](#). In `src/`:

```
cc -O2 -frounding-math -ffp-contract=off -o rigcert rigcert.c -lm # GNU gcc: -O0
K=50:0.07,80:0.1612,130:0.2191,220:0.2679,400:0.3198,800:0.345
K=$K,2000:0.3762,8000:0.4031,50000:0.4092
KNOTS=$K PSB=31 DSB=0.05 NOSTOP=1 \
./rigcert 16000 50 200000 2e8 47 16 4.5 1000000 20000 0 -1 0.1 0.44 0.5
```

The arguments are, in order: m ; the prime 50 from which the knot schedule applies; the largest prime $2 \cdot 10^5$ treated by the comparison bound; the largest prime $2 \cdot 10^8$ treated explicitly; the largest prime 47 allowed in s_A ; the constants 16 and 4.5 of the enumeration cut-off X_p ; the caps 10^6 and $2 \cdot 10^4$ of the divisor-count tables; a verbosity flag; and an unused smooth schedule (ignored when `KNOTS` is set). `PSB` and `DSB` give $\delta_p = 0.05$ for $31 \leq p < 50$; `NOSTOP` only disables an early exit. The expected output (`logs/cert16k.txt`; line breaks added) is

```
m=16000 eta_up=0.997384393974 (A=0.230571210247 B=0.760302397508 C=0.006510785610)
f_up=1.22745e+08 crit_lo=2.83863e+09 SUCCESS(certified)
CERT eta_up=0.9973843939732947 (0x1.fea92ad34fe9cp-1) T_up=321051.36453405279
mu_lo=0.0026156060267052972 f_up=122744542.28049764 crit_lo=2838631802.0518227
k=11078937 log_k_lo=16.220556296052827 loglog_k_lo=2.7862793450197798
CERT stats: max_enumerated=12114518 t_clamped_to_delta=0
```

With `DUMP=file` the program writes the certified per-prime costs for $p \leq 2 \cdot 10^5$ (`logs/dump16k.txt`). The run takes about 2.5 minutes on one core. The commands of the independent re-computation are listed at the end of `logs/xcheck_final.md` (code in `tests/xcheck_final/`); `tests/cert/run_small.sh` runs the small-instance checks of [Section 6.6](#), and the tests of the comparison theorem are the Python scripts in `tests/review/`.

This paper. In `paper/`, `make_figures.py` regenerates `fig/` and `gen/` from the certified outputs, and `tectonic main.tex` builds the paper.

Lean. In the root of the Lean development:

```
lake exe cache get      # prebuilt Mathlib
lake build              # Theorems 1.1 and 1.3 (native checks: about 10 min and 50 s)
lake build MinModulus.Audit  # axioms of both theorems
```